MCMC for Bayesian Clustering with a Random Number of Components

Stochastic block models and Gaussian mixtures with a prior on K: what is known about convergence, and specific versions of project P3

Section 1 Annotated literature: qualitative and quantitative convergence results
Section 2 Specific research projects, with heuristics and numerical checks

Literature checked up to 19 October 2026

Badges. [Q0] invariance only [Q1] plus irreducibility argued [Q2] ergodicity proved [Q3] quantitative bound **[F]** failure [tool] general theory [context] background.

Verification. FT full text read Abs abstract or summary Sec secondary source Meta bibliographic metadata only † a detail is unverified.

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Overview

Question. For Gaussian mixtures and stochastic block models in which the number of components K is random with a prior, what has been proved about the convergence of MCMC samplers? How does general Markov chain theory deliver qualitative convergence when K is random, and has there been real progress on quantitative bounds? Section 1 answers these questions from the literature. Section 2 turns the gaps into specific projects.

Main findings of Section 1.

1. *Qualitative convergence comes almost for free, but only in weak forms.* The samplers people use on partitions have a finite state space, so irreducibility and aperiodicity already give uniform ergodicity, whatever the

prior on K. Samplers that carry K and continuous parameters (reversible jump, birth–death) rely on Tierney's theorem, which gives convergence from almost every starting point. Geometric ergodicity has been proved for mixtures with random K only when K is bounded (T13). A second case, a Dirichlet-process-based model with at most M components, is claimed as a "minor modification" of a fixed-K proof (M13). No sampler for an SBM with random K has a proved ergodicity theorem beyond the automatic finite-state one.

2. *There is no quantitative result with K random.* No mixing time, relaxation time or spectral gap bound, upper or lower, was found for any sampler targeting a Gaussian-mixture or SBM posterior with random K. The closest are Qin's decomposition of the spectral gap into within- and between-model parts (T13), a 2026 extension of the Ascolani–Zanella variance comparison to random partitions that is announced but was not seen (R1), and coupling bounds that degenerate with n (D10).

3. *With K known, there are a few sharp results.* Zhuo & Gao (U1) prove the only rapid-mixing theorem for a Bayesian SBM posterior. Tosh & Dasgupta (L1) prove lower bounds for collapsed Gibbs on Gaussian mixtures. They are exponential in the separation when k is misspecified, and of order $n^{\Omega(\alpha)}$ for the correct k. Mou et al. (U2) prove polynomial mixing for a tempered two-component posterior.

4. *Practice agrees that moves which change K are the bottleneck* (S9, S7, M8, M9), but no theorem says when or why.

Main proposals of Section 2. The flagship project (P3.A) is an exponential-in-dimension barrier for single-site Gibbs samplers in high-dimensional Gaussian mixtures with a prior on K. Single-site moves can open a missing cluster only if the separation Δ^2 is of order p. Yet the posterior should already concentrate on the right K once $\Delta^2 \gtrsim (pK/n)\log n$, and split–merge moves find it at once. New simulations (Section 2.3) support this window. Over-clustered starts that give each cluster a large block avoid the barrier. Exact profile computations (Section 2.8) suggest that tempering raises the barrier rather than lowering it. The same mechanism appears in sparse multilayer SBMs, with the number of layers in the role of the dimension (P3.B, linked to P4). It also appears in ordinary SBMs, with the number of communities in that role (P3.C). There, opening a community costs a factor of about $n^{-K/2}$, which suggests why single-site samplers underestimate K when K is large. Three further projects are complementary: how the tail of the prior on K decides geometric ergodicity (P3.D, the lowest-risk one), posterior scaling limits in low dimension (P3.E), and torpid mixing of tempering (P3.F).

1 Literature: MCMC convergence for SBMs and Gaussian mixtures with a random K

1.1 Scope, method and legend

Inclusion. An article is included when it meets all four conditions below.

- It concerns a Gaussian mixture model or a stochastic block model. Close relatives are included: other location mixtures, degree-corrected and weighted SBMs, infinite relational and latent block models.
- K is random with a prior. This covers mixtures of finite mixtures (MFM), Dirichlet- and Pitman–Yor-process mixtures and other random-partition priors (where the number of clusters is random), and trans-dimensional formulations.
- An MCMC method is used, of any kind: Gibbs, Metropolis–Hastings, reversible jump, birth–death, split–merge, slice, coupled or tempered.
- The article makes a statement about the convergence of the sampler, from invariance to explicit rates.

As requested, four kinds of related work are also included:

- the general Markov chain theory these papers rely on;
- quantitative results with K known;
- the nearest quantitative analogues: chains on partitions without data, trans-dimensional samplers outside clustering, tempering, and lower-bound templates from planted problems;
- a few papers whose MCMC has no theory but which document its convergence behaviour, plus the statistical results that say what the samplers target.

Levels.

- **[Q0]** The sampler has the right stationary distribution by construction (detailed balance or exact Gibbs steps), and nothing more is claimed.
- **[Q1]** As **[Q0]**, plus irreducibility or aperiodicity is argued or sketched.
- **[Q2]** An ergodicity theorem is proved: convergence in total variation from almost every or every start, Harris recurrence, or uniform or geometric ergodicity.
- **[Q3]** A quantitative statement is proved: a mixing, relaxation or hitting time, a spectral-gap bound, or a lower bound of this kind.
- **[F]** Failure is documented: the chain cannot converge, or provably does not converge, to a probability distribution.

Verification. FT means the full text (usually the arXiv version) was read. Abs means the abstract or a summary was read. Sec means the content comes from a secondary source, such as a citing paper. Meta means only the bibliographic metadata was checked. A dagger † marks a bibliographic or technical detail that could not be checked.

Search. Searches and reading were done between 15 and 19 October 2026, organised by topic: Gaussian mixtures with random K; SBMs with random K; quantitative mixing results; and a sweep of 2021–2026. Two limits apply.

- The search budget ran out before some planned queries were run.
- Project Euclid, Taylor & Francis, ScienceDirect and PMC pages could not be opened. For papers published there, the arXiv version and the metadata were used.

"Not found" therefore means not found in a partial search. Entries marked *(our observation)* are remarks made for this document. They are not claims taken from the cited papers.

1.2 Summary: what is known

How general theory delivers qualitative convergence when K is random

Random K needs no special theory. Which result applies depends on the state space of the sampler, not on the model, and there are four routes.

Route 1: a finite state space. Collapsed samplers integrate out the weights, the component parameters and K itself. They then move on the set of partitions of $\{1, \dots, n\}$, which is finite (a Bell number of partitions), whatever the prior on K, because the number of clusters satisfies $K_+ \leq n$.

- Examples, all with conjugate components so that the parameters can be integrated out, are the MFM Gibbs sampler (M8), Neal's Algorithm 3 for Dirichlet-process mixtures (D1), conjugate split–merge (D2), and the collapsed SBM samplers (S9, S7, S13). Samplers with auxiliary parameters, such as Neal's Algorithm 8, are not finite-state.
- A finite chain that is irreducible and aperiodic is uniformly ergodic. This is a textbook fact, and irreducibility is immediate when single-site moves connect all partitions of positive posterior mass.
- So for the most widely used samplers **[Q2]** is automatic, and every open question is quantitative. The papers seldom say this. The rate it gives can be as bad as $1 - e^{-cn}$ per step.

Route 2: a countable state space. Allocation samplers keep K explicit and allow empty components (M5, S3, S12). With an unbounded prior on K, their state space is countable.

- An irreducible, aperiodic chain with an invariant probability is positive recurrent, so it converges from every state.
- Geometric ergodicity is *not* automatic. Section 2.6 observes that the posterior of K inherits the tail of the prior. If that tail is polynomial (beta-negative-binomial or Gnedin-type priors), a sampler that changes K by bounded jumps cannot be geometrically ergodic, by the exponential-moment criterion (T19) *(our observation)*.

Route 3: a general state space. Reversible jump, birth–death and conditional samplers keep continuous weights and parameters.

- Invariance is built in (T12, M2, M3). Tierney's theorem then gives convergence from π-almost every starting point under φ-irreducibility and aperiodicity (T1, T2).
- Convergence from *every* start needs Harris recurrence. This can fail for φ-irreducible chains, and Roberts & Rosenthal give sufficient conditions for trans-dimensional chains (T3). Nobody has checked these conditions for mixture samplers.
- Geometric ergodicity has been proved in one case: Qin's reversible jump for normal mixtures with $K \leq k_{\max}$ (T13). A second case, uniform ergodicity for a Dirichlet-process-based mixture with at most M components, is claimed to follow "with minor modifications" from a fixed-K proof (M13).
- Nothing has been proved for unbounded K, for split–merge reversible jump, or for Stephens' birth–death sampler. For the last one, only invariance is proved, and the classical convergence conditions for spatial birth–death processes need death rates bounded below (T10), which Stephens' likelihood-ratio rates are not *(inference, not checked in print)*.

Route 4: duality. Suppose a sampler alternates between a finite discrete block (allocations or a partition) and the rest, and the rest is drawn exactly from its full conditional. Then the discrete block is a finite Markov chain, and the whole chain converges as fast as it does (T7, T6, T8). The telescoping sampler (M9) with conjugate components and fixed hyperparameters seems to have this form, because K and the empty components are redrawn exactly given the partition. It would then be uniformly ergodic for *any* prior on K *(our observation; see P3.D)*.

Failures. With Jeffreys priors the posterior can be improper, so no chain converges to a probability distribution (M14). There are also φ-irreducible chains that fail to converge from uncountably many starts (T3, Example 9).

Assumptions. Qualitative results use no separation, balance or signal assumptions of the kind in P1. They use positivity of the target, bounded or finite state spaces, and integrability. The quantitative results with K known (U1, L1) are where separation and balance enter.

Sampler class	Examples	State space	Automatic from general theory	Proved beyond that
Collapsed, on partitions	M8, D1, D2, S9, S7, S13	finite	uniform ergodicity once irreducible and aperiodic	nothing quantitative
Allocation samplers with explicit K and empty components	M5, S3, S12, S8	countable; finite if K is bounded (e.g. $k \leq n$ in S8)	convergence from every state	no geometric ergodicity proof; fails for heavy-tailed p_K (P3.D)
Conditional, with K drawn given the partition	M9, M10, M11, D5	general	convergence from π-a.e. start; duality may reduce it to a finite chain	none
Reversible jump, birth–death	M1, M2, M3, S11	general, a union over k	invariance; π-a.e. convergence	geometric ergodicity for $K \leq k_{\max}$ only (T13); M13
Coupled or perfect	D10, M13	—	unbiased estimators; CFTP terminates iff uniformly ergodic	meeting-time bounds degenerate in n

Table 1: How qualitative convergence is obtained for random-K samplers, by class of sampler.

Progress on quantitative results

With K random. No mixing-time, relaxation-time or spectral-gap bound, upper or lower, was found for any sampler that targets a Gaussian-mixture or SBM posterior with random K. The partial results are these:

- *A decomposition.* Qin's Theorem 7 (T13) bounds the spectral gap of a trans-dimensional chain by three factors: a lower bound on the within-model mixing weights, the worst within-model gap, and the gap of the between-model chain. It has not been evaluated for any mixture.
- *Comparisons.* The 2026 version of Ascolani–Manildo–Zanella (R1) announces an extension to random partitions of an asymptotic-variance comparison between collapsed Gibbs and a non-reversible sampler. The theorem is in a supplement that was not seen. Their scaling limits remain for fixed K and for the prior only.
- *Couplings and cost.* Coupled chains on partitions (D10) come with meeting-time bounds that degenerate with n. Franzolini & Gaffi (D11) bound the per-iteration cost of slice samplers, not their mixing.
- *Idealised dynamics.* Scaling limits exist for idealised model-index dynamics (R2, R3).
- *The prior alone.* Without data, the random-scan single-site collapsed sampler for a Chinese restaurant process is the infinite-alleles Moran model of population genetics *(our identification)*. A cutoff is known for that model when the mutation rate grows with n (P2). For fixed α the time scale is $\Theta(n^2)$ updates.

With K known. There are sharp results:

- *Upper bounds.* Rapid mixing of Metropolis–Hastings for the SBM posterior from a warm start above the exact-recovery threshold (U1). Polynomial mixing for a tempered two-component posterior (U2). Simulated-tempering samplers for two-cluster label posteriors (U3). Conditional mixing of Langevin dynamics (U4).
- *Lower bounds.* Lower bounds for collapsed Gibbs on Gaussian mixtures, exponential in the separation when k is misspecified and of order $n^{\Omega(\alpha)}$ for the correct k (L1). Exponential lower bounds on two-component mixtures (L2) and on motif mixtures (L3). Exponential slow mixing for a modularity-tilted Gibbs measure on the SBM (L4).
- *SBM Glauber dynamics.* Weak recovery at a random time, which falls short of mixing (U6).

Bottom line. Real progress consists of:

- one rapid-mixing theorem for a Bayesian clustering posterior used in practice, with K known (U1);
- a decomposition that reduces trans-dimensional rates to within- and between-model rates (T13);
- comparison and scaling-limit techniques that have just begun to reach random partitions (R1).

No theorem explains when the moves that change K are slow. That is the gap Section 2 targets.

Setting	Upper bounds (fast mixing)	Lower bounds (slow mixing)
Gaussian mixture, K random	none found	none found (heuristics and simulations: P3.A)
Gaussian mixture, K known	U2 (tempered, $K = 2$); U3 (simulated tempering, $K = 2$); U4 (conditional); U5 (constrained Langevin)	L1 (collapsed Gibbs); L2 (two components); L3 (motif mixtures); T8 (numerical eigenvalues)
SBM, K random	none found	none found (heuristics and simulations: P3.B, P3.C)
SBM, K known	U1 (warm start); U6, U7 (Glauber, surrogate measures); U9 (synchronisation analogue)	L4 (modularity measure); heuristic H8
Partitions without data	P2 (random number of blocks); P1, P4, P8, P9	—
Trans-dimensional, outside clustering	V1, V2, V3, V4	V6, V7

Table 2: Quantitative results at a glance.

1.3 General Markov chain theory used when K is random

T1 Markov chains for exploring posterior distributions (with discussion) [tool] Sec

L. Tierney (1994). *Ann. Statist. 22(4), 1701–1762*

The basic convergence theory of MCMC. A φ-irreducible, aperiodic chain with invariant law π converges in total variation from π-almost every starting point. This is the theorem behind nearly every convergence claim for reversible-jump and conditional mixture samplers. Roberts & Rosenthal (T3) cite it for Harris recurrence of φ-irreducible full-dimensional Metropolis–Hastings and Gibbs samplers.

T2 General state space Markov chains and MCMC algorithms [tool] FT

G. O. Roberts, J. S. Rosenthal (2004). *Probab. Surveys 1, 20–71. doi: 10.1214/154957804100000024*

A survey that states the main tools.

- Theorem 4: φ-irreducibility, aperiodicity and an invariant π give convergence from π-a.e. start, and Harris recurrence gives convergence from every start.
- Theorem 8: a minorisation on the whole space gives uniform ergodicity.
- Theorem 9: a drift condition with a small-set minorisation gives geometric ergodicity.

There is no trans-dimensional material.

T3 Harris recurrence of Metropolis-within-Gibbs and trans-dimensional Markov chains [tool] FT

G. O. Roberts, J. S. Rosenthal (2006). *Ann. Appl. Probab. 16(4), 2123–2139. doi: 10.1214/105051606000000510*

- Example 9: a φ-irreducible, aperiodic two-dimensional Metropolis-within-Gibbs chain that fails to converge from uncountably many starting points.
- Theorem 20: a reversible-jump chain whose within-model moves are full-dimensional Metropolis–Hastings is Harris recurrent if, from every start, a within-model move is eventually accepted almost surely. The between-model moves can be anything that preserves π.
- Theorem 21: a φ-irreducible coordinate-preserving chain that eventually moves every coordinate is Harris recurrent.

Open problems are listed. Mixtures are not discussed.

T4 Two convergence properties of hybrid samplers [tool] FT (preprint)

G. O. Roberts, J. S. Rosenthal (1998). *Ann. Appl. Probab. 8(2)†. doi: 10.1214/aoap/1028903533*

Random-scan hybrid samplers inherit uniform ergodicity from their components, with explicit but crude constants. Random-walk Metropolis-within-Gibbs is geometrically ergodic under tail conditions.

T5 Component-wise Markov chain Monte Carlo: uniform and geometric ergodicity under mixing and composition [tool] Abs

A. A. Johnson, G. L. Jones, R. C. Neath (2013). *Statist. Sci. 28(3), 360–375. doi:10.1214/13-STS423*

Relates the convergence rates of deterministic-scan, random-sequence and random-scan samplers. Also gives conditions under which component-wise samplers with state-independent proposals are uniformly ergodic.

T6 Markov chains and de-initializing processes [tool] Meta

G. O. Roberts, J. S. Rosenthal (2001). *Scand. J. Statist. 28(3), 489–504*

If one process "de-initialises" another, the second is at least as close to stationarity as the first. Given X_t, the future of Y must not depend on the initial state. This is the formal version of the duality principle for two-block Gibbs samplers, used in P3.D. The statement is the authors' well-known result; only the metadata was re-checked here.

T7 Estimation of finite mixture distributions through Bayesian sampling [K known] [Q2] Abs Sec

J. Diebolt, C. P. Robert (1994). *J. R. Stat. Soc. Ser. B 56(2)†, 363–375*

Data augmentation for mixtures with K fixed. Convergence follows from a duality principle: properties of the finite allocation chain carry over to the parameter chain. The sampler is ergodic, and geometrically so in the normal case. Later work by Lee, Marin, Mengersen & Robert restates that the fixed-k Gibbs sampler is uniformly geometrically ergodic, yet it suffers from trapping states.

T8 Improving the convergence properties of the data augmentation algorithm with an application to Bayesian mixture modeling [K known] [Q2] FT

J. P. Hobert, V. Roy, C. P. Robert (2011). *Statist. Sci. 26(3), 332–351. doi:10.1214/11-STS365; arXiv:0911.4546*

With a finite augmentation space (k^m allocations), the data-augmentation operator is compact with finite spectrum in $[0, 1)$, so the chain is uniformly ergodic. The sandwich chain that adds a random label permutation (Frühwirth-Schnatter's random permutation sampler) has eigenvalues no larger than plain data augmentation. In a two-component Bernoulli toy model, the top data-augmentation eigenvalue is 0.99395 at $m = 10$ and 0.99996 at $m = 20$, against 0.198 and 0.152 with the permutation step. For normal mixtures, the paper says there are "no theoretical results" on the eigenvalues.

T9 Spatial birth-and-death processes [tool] Sec

C. Preston (1976/77). *Bull. Int. Statist. Inst. 46, 371–391*

Proves existence, uniqueness and convergence of spatial birth–death processes. The process is coupled with a one-dimensional birth–death process whose rates are the supremum of the birth rates and the infimum of the death rates over n-point configurations. Local balance identifies the invariant law, and this is the part Stephens (M2) uses. The year is cited as both 1976 and 1977.

T10 On the rate of convergence of spatial birth-and-death processes [tool] FT

J. Møller (1989). *Ann. Inst. Statist. Math. 41(3), 565–581*

Proves geometric convergence in total variation, $\|Q_t - \nu\| \le c\, r^t$, under three conditions: bounded birth rates, death rates bounded below over n-point configurations, and a summability condition. In the truncated case the constants do not depend on the start.

T11 Simulation procedures and likelihood inference for spatial point processes [tool] Sec

C. J. Geyer, J. Møller (1994). *Scand. J. Statist. 21(4), 359–373*

Birth–death Metropolis–Hastings for point processes, a discrete-time ancestor of reversible-jump birth and death moves. Qin (T13) credits it with proving geometric ergodicity of a trans-dimensional algorithm. The exact theorem and its conditions were not checked.

T12 Reversible jump Markov chain Monte Carlo computation and Bayesian model determination [Q0] Abs Sec

P. J. Green (1995). *Biometrika 82(4), 711–732. doi:10.1093/biomet/82.4.711*

The reversible-jump framework. Detailed balance follows from how the acceptance ratio is constructed. Irreducibility and aperiodicity are left to each application.

T13 Geometric ergodicity of trans-dimensional Markov chain Monte Carlo algorithms [K random] [Q2] FT

Q. Qin (2025). *J. Amer. Statist. Assoc. 120(551), 1868–1878 (online 2024). doi:10.1080/01621459.2024.2427432; arXiv:2308.00139*

The only fully proved geometric-ergodicity theorem found for a Gaussian-mixture sampler with

random K (compare M13).

- *Theorem 1* assumes finitely many models and three conditions. Each within-model kernel is L^2-geometrically convergent on its model. The full kernel dominates it, with weight c_k, inside model k. The between-model chain is irreducible. Then the trans-dimensional chain is $L^2(\Pi)$-geometrically convergent. Reversibility is not needed.
- *Theorem 7* is a quantitative version of Madras–Randall type that allows countably many models. The gap of the full chain is bounded below by a product of $\inf_k c_k$, the worst within-model gap and the gap of the averaged between-model chain. The two reports read differ on the exact powers.
- *Proposition 13* treats a univariate normal mixture with $K \in \{1, \ldots, k_{\max}\}$ ($k_{\max} = 30$ for the galaxy data). The priors are Dirichlet weights, ordered normal means and inverse-gamma variances. The moves are birth of a new component, death of an empty one, "Metropolis re-ordering" and Gibbs updates. The chain is L^2-geometrically convergent and Π-a.e. geometrically ergodic. No rate is given, and nothing is said about the dependence on n. The key step is that the within-model kernel depends on the state only through the finite allocation vector.

Split–merge moves are only suggested (Remark 14). Sharp rates are called "an open problem for most practical trans-dimensional MCMC algorithms".

T14 Solidarity of Gibbs samplers: the spectral gap [tool] Abs Sec

I. Chlebicka, K. Łatuszyński, B. Miasojedow (2025). *Ann. Appl. Probab. 35, 142–157*†*; arXiv: 2304.02109*

If one random-scan or deterministic-scan Gibbs sampler has a spectral gap, they all do.

T15 Extensions of the solidarity principle of the spectral gap for Gibbs samplers to their blocked and collapsed variants [tool] FT

X. Mak, J. P. Hobert (2026). *arXiv: 2601.06745*

Blocked and collapsed variants inherit a spectral gap from a full Gibbs sampler, and enlarging blocks cannot hurt. One example shows that a variant need not inherit geometric ergodicity from another. The results are qualitative and assume finitely many coordinates, so they apply to collapsed finite-mixture samplers but not to Dirichlet-process samplers.

T16 Spectral gap bounds for reversible hybrid Gibbs chains [tool] Abs

Q. Qin, N. Ju, G. Wang (2023). *arXiv: 2312.12782*

Bounds the spectral gap of Metropolis-within-Gibbs and hybrid slice samplers by that of the exact Gibbs sampler and the gaps of the inner chains.

T17 Approximate spectral gaps for Markov chains mixing times in high dimensions [tool] FT

Y. F. Atchadé (2021). *SIAM J. Math. Data Sci. 3(3), 854–872*†*; arXiv: 1903.11745*

Defines an approximate (ζ-) spectral gap for chains whose true gap is near zero because of tiny regions of the space. For Gibbs samplers on mixtures of densities it sharpens Madras–Randall decompositions. It is applied to variable selection, where it gives polynomial mixing from a good initial distribution. The two reports read identify the published version differently, so that detail is flagged.

T18 Geometric ergodicity of Metropolis algorithms [tool] Abs

S. F. Jarner, E. Hansen (2000). *Stochastic Process. Appl. 85(2), 341–361*

For random-walk Metropolis, target tails that are at least exponentially light are necessary for geometric ergodicity. This extends the one-dimensional result of Mengersen & Tweedie (1996).

T19 Necessary conditions for geometric and polynomial ergodicity of random-walk-type Markov chains [tool] Meta

S. F. Jarner, R. L. Tweedie (2003). *Bernoulli 9(4), 559–578*†*. doi: 10.3150/bj/1066223269*

For chains whose increments have uniformly bounded exponential moments, geometric ergodicity forces the target to have exponential moments. P3.D uses this to show that a sampler changing K by bounded jumps is never geometrically ergodic when the posterior of K has polynomial tails. The statement comes from the authors' known result; only the metadata was checked.

T20 Mixing times are hitting times of large sets [tool] Meta

Y. Peres, P. Sousi (2015). *J. Theoret. Probab. 28(2), 488–519*†*. doi: 10.1007/s10959-013-0497-9*

For lazy reversible chains, the mixing time is equivalent, up to constants, to the largest expected hitting time of a set with stationary mass at least α, for any $\alpha < 1/2$. Oliveira (2012, arXiv: 1108.1708)

proved a similar result independently. Section 2 uses this to turn hitting-time bounds into mixing bounds when the posterior concentrates.

1.4 Finite mixtures with a prior on K (Gaussian and related)

M1 On Bayesian analysis of mixtures with an unknown number of components (with discussion) [K random] [Q1] FT

S. Richardson, P. J. Green (1997). *J. R. Stat. Soc. Ser. B 59(4), 731–792*

Univariate Gaussian mixture, $K \sim \text{Uniform}\{1, \ldots, 30\}$, with proper hierarchical priors. The sampler is reversible jump with split/combine moves and birth/death of empty components, plus Gibbs updates within K. Section 3.2 notes that detailed balance holds and that "it only remains to check" irreducibility and aperiodicity, which it argues in a few lines. There is no ergodicity theorem, and convergence is assessed empirically.

M2 Bayesian analysis of mixture models with an unknown number of components—an alternative to reversible jump methods [K random] [Q0]/[Q1] FT (preprint)

M. Stephens (2000). *Ann. Statist. 28(1), 40–74. doi:10.1214/aos/1016120364*

Normal mixtures with a truncated Poisson prior on k ($k_{\max} = 100$). Continuous-time birth–death MCMC alternates with fixed-k Gibbs. Theorem 3.1 shows that the posterior is stationary, by checking Preston's balance conditions (T9). Irreducibility of the combined chain is asserted "provided the full conditionals give support to all parts of the parameter space", with the proof in the author's thesis. There is no ergodicity or rate result. The convergence theorems for spatial birth–death processes (T10) need death rates bounded below, while Stephens' death rates are likelihood ratios that can be arbitrarily small *(inference, not checked in print)*.

M3 Reversible jump, birth-and-death and more general continuous time Markov chain Monte Carlo samplers [K random] [Q0] FT

O. Cappé, C. P. Robert, T. Rydén (2003). *J. R. Stat. Soc. Ser. B 65(3), 679–700*

Gaussian mixtures and hidden Markov models with random K. Theorem 1 shows that the time-rescaled discrete reversible-jump chain converges weakly, in the Skorokhod space, to the birth–death process. The assumptions are a Polish parameter space, positive continuous birth rate, prior and likelihood, and a locally integrable proposal. Invariance follows from local balance. Non-explosion and ergodicity are not proved. Empirically, reversible jump and birth–death are about equally accurate, and reversible jump is cheaper.

M4 On the posterior distribution of the number of components in a finite mixture [K random] [context] Abs

A. Nobile (2004). *Ann. Statist. 32(5)†. doi:10.1214/009053604000000788*

Not about convergence. Derives inequality constraints on the posterior of k that MCMC estimates must satisfy, which can serve as a diagnostic.

M5 Bayesian finite mixtures with an unknown number of components: the allocation sampler [K random] [Q0] Abs

A. Nobile, A. T. Fearnside (2007). *Statist. Comput. 17(2), 147–162. doi:10.1007/s11222-006-9014-7*

A collapsed sampler on (number of components, allocations), with conjugate parameters integrated out. Fearnside's thesis truncates a Poisson(1) prior at $k_{\max}$. No convergence claim appears in the abstract. With a truncated prior the state space is finite, so irreducibility would immediately give uniform ergodicity *(our observation)*.

M6 Further reversible-jump samplers for normal mixtures [K random] [Q0] Meta

P. Dellaportas, I. Papageorgiou (2006); Z. Zhang, K. L. Chan, Y. Wu, C. Chen (2004); P. Papastamoulis, G. Iliopoulos (2009) (2004–2009). *Statist. Comput. 16(1), 57–68 (doi:10.1007/s11222-006-5338-6); Statist. Comput. 14, 343–355 (doi:10.1023/B:STCO.0000039484.36470.41); Comput. Statist. Data Anal. 53(4), 900–911*

Univariate and multivariate Gaussian mixtures with split/combine and birth/death moves. The work is about proposal design, and no convergence theory was seen. The second paper is *Learning a multivariate Gaussian mixture model with the reversible jump MCMC algorithm*. The other two titles were not re-checked.

M7 Markov chain Monte Carlo methods and the label switching problem in Bayesian mixture model-

ing [context] FT

A. Jasra, C. C. Holmes, D. A. Stephens (2005). *Statist. Sci. 20(1), 50–67. doi: 10.1214/088342305000000016*

A review with no theorems. It treats visiting all $k!$ labellings as necessary for convergence, and the failure to switch labels as evidence of non-convergence.

M8 Mixture models with a prior on the number of components [K random] [Q1] FT

J. W. Miller, M. T. Harrison (2018). *J. Amer. Statist. Assoc. 113(521), 340–356. doi: 10.1080/01621459.2016.1255636; arXiv: 1502.06241*

Mixtures of finite mixtures (MFM) with $K \sim p_K$, possibly unbounded. The samplers are the partition-based collapsed Gibbs sampler (Neal's Algorithm 3 with MFM weights), Algorithm 8 and Jain–Neal split–merge. Section 6 notes that the chain must be irreducible to be valid and argues that it is, because $V_n(t) > 0$. For the conjugate partition-based sampler the state space is finite, so uniform ergodicity follows *(our observation)*. Algorithm 8 carries auxiliary parameters and is not covered by this argument. Empirically, incremental Gibbs mixes more slowly for MFMs than for DP mixtures.

M9 Generalized mixtures of finite mixtures and telescoping sampling [K random] [Q0] FT

S. Frühwirth-Schnatter, G. Malsiner-Walli, B. Grün (2021). *Bayesian Anal. 16(4)†. doi: 10.1214/21-BA1294; arXiv: 2005.09918*

K is unbounded in theory, with $K_{\max} = 100$ in practice. The recommended prior on K is the beta-negative-binomial $\mathrm{BNB}(1, 4, 3)^{\dagger}$ (as recalled from the paper; not re-checked), which has polynomial tails. The telescoping sampler draws K from $p(K \mid \text{partition})$. Only the validity of that step is shown. The authors warn that mixing is poor when $p(K \mid \text{partition})$ concentrates at K_+. For conjugate components with fixed hyperparameters, the sampler appears to be a two-block Gibbs sampler, partition versus the rest, which would make it uniformly ergodic by duality *(our observation; P3.D)*.

M10 Is infinity that far? A Bayesian nonparametric perspective of finite mixture models [K random] [Q0] FT

R. Argiento, M. De Iorio (2022). *Ann. Statist. 50(5), 2641–2663. doi: 10.1214/22-AOS2201; arXiv: 1904.09733*

Normalised independent finite point processes, with $M \sim q_M$ on the positive integers, unbounded. A conditional Gibbs sampler with a latent variable u updates the number of non-allocated components. Validity is said to "follow directly from Theorem 2". Comparisons with reversible jump are empirical (integrated autocorrelation times and effective sample sizes).

M11 Gibbs sampling for mixtures in order of appearance: the ordered allocation sampler [K random] [Q0] FT

P. De Blasi, M. F. Gil-Leyva (2023). *J. Comput. Graph. Statist. 32(4), 1416–1424. doi: 10.1080/10618600.2023.2177298; arXiv: 2107.08380*

A random dimension m with a prior. Gibbs plus Metropolis–Hastings, with m updated in a block. There are no chain-level results, and only empirical comparisons of integrated autocorrelation times.

M12 Fast sampling and model selection for Bayesian mixture models [K random] [Q1] FT

M. E. J. Newman (2025). *Statist. Comput. (2025)†. doi: 10.1007/s11222-025-10753-0; arXiv: 2501.07668*

Latent class models with K random (uniform prior). The sampler is rejection-free on the collapsed posterior $P(k, z \mid x)$ and picks a component first. Detailed balance is proved, including for the moves that change K, and ergodicity is argued briefly. A heuristic suggests that components are removed $O(N/k)$ times faster than with standard moves. Otherwise the evidence is empirical autocorrelation times.

M13 Perfect simulation for mixtures with known and unknown number of components [K random] [Q2] FT

S. Mukhopadhyay, S. Bhattacharya (2012). *Bayesian Anal. 7(3), 675–714. doi: 10.1214/12-BA723; arXiv: 1102.4152*

Normal mixtures. For unknown K the paper uses Bhattacharya's Dirichlet-process-based model with a known maximum M, so the number of distinct components ($\leq M$) is random. The method is Gibbs sampling plus coupling from the past with bounding chains. Uniform ergodicity of the Gibbs sampler is shown by minorisation (supplement S-9). The authors say the proof extends to unknown K "with minor modifications", and pair it with Foss–Tweedie: coupling from the past terminates exactly when the chain is uniformly ergodic. The non-conjugate case needs a compact parameter space. The S-9 proof was not checked independently.

M14 Jeffreys priors for mixture estimation [K known] [F] FT

C. Grazian, C. P. Robert (2018). *arXiv: 1511.03145; journal version not checked*

The posterior is improper under Jeffreys priors in three cases: all parameters unknown (Theorem 3.1), only the scales unknown (Lemma 3.7), and only the means unknown with $k > 2$ (Lemma 3.6). MCMC output "is not conclusive" for detecting this, because chains can look stable or get stuck. No chain can converge to a probability distribution.

M15 Computational and inferential difficulties with mixture posterior distributions [K known] [context] Abs

G. Celeux, M. Hurn, C. P. Robert (2000). *J. Amer. Statist. Assoc. 95†, 957–970. doi: 10.2307/2669477*

Samplers get trapped in modes. The paper discusses tempered transitions and loss functions for label switching. It is qualitative only.

M16 Blocked Gibbs sampling for improved convergence in finite mixture models [K known] [context] Abs

D. M. Swanson (2024). *arXiv: 2411.00371*

Outliers roughly equidistant between components can make allocations highly correlated. Rate bounds are proved only for a modified stationary distribution, and the gain from blocking is empirical.

1.5 Dirichlet- and Pitman–Yor-process mixtures

D1 Markov chain sampling methods for Dirichlet process mixture models [K random] [Q1] FT

R. M. Neal (2000). *J. Comput. Graph. Statist. 9(2), 249–265*

Every update leaves the posterior invariant. Ergodicity is asserted for Algorithm 1, with a warning that it can be slow, and for Algorithm 7, since one scan can put every observation in its own cluster with positive probability. Algorithm 8 is exact for any number m of auxiliary components. There are no rates.

D2 A split-merge Markov chain Monte Carlo procedure for the Dirichlet process mixture model [K random] [Q0] Abs

S. Jain, R. M. Neal (2004). *J. Comput. Graph. Statist. 13(1)†, 158–182. doi: 10.1198/1061860043001*

Split–merge Metropolis–Hastings for conjugate DP mixtures. The split proposal is built from a launch state refined by restricted Gibbs scans. The motivation is that single-site samplers rarely create or remove clusters.

D3 Splitting and merging components of a nonconjugate Dirichlet process mixture model [K random] [Q0] Meta

S. Jain, R. M. Neal (2007). *Bayesian Anal. 2(3), 445–472†*

The non-conjugate extension of D2.

D4 Retrospective Markov chain Monte Carlo methods for Dirichlet process hierarchical models [K random] [Q0] FT

O. Papaspiliopoulos, G. O. Roberts (2008). *Biometrika 95(1), 169–186; arXiv: 0710.4228*

There is no geometric-ergodicity proof. The exact allocation step needs a bounded likelihood. The Mengersen–Tweedie criterion for independence samplers is used only as a tuning heuristic. Label-switching moves are shown empirically to be "crucial".

D5 Sampling the Dirichlet mixture model with slices; Slice sampling mixture models [K random] [Q0] FT (Kalli et al.) Sec (Walker)

S. G. Walker (2007); M. Kalli, J. E. Griffin, S. G. Walker (2011) (2007, 2011). *Commun. Statist. Simul. Comput. 36(1), 45–54; Statist. Comput. 21, 93–105 (doi: 10.1007/s11222-009-9150-y)*

Slice samplers are "a valid Markov chain with correct stationary distribution", because each iteration samples a finite set of variables. There are no ergodicity statements, only comparisons of integrated autocorrelation times.

D6 Parallel sampling of DP mixture models using sub-cluster splits [K random] [Q0] Abs

J. Chang, J. W. Fisher III (2013). *Adv. Neural Inf. Process. Syst. 26 (NIPS 2013)*

A "non-ergodic restricted Gibbs iteration" combined with split–merge proposals generated by sub-clusters attached to each cluster. The combination is valid, and convergence is reported to be better

empirically. This is practical evidence that cluster-creating moves must be designed separately.

D7 Sequentially allocated merge-split samplers for conjugate Bayesian nonparametric models [K random] [Q0] Meta

D. B. Dahl, S. Newcomb (2022). *J. Statist. Comput. Simul.* doi: 10.1080/00949655.2021.1998502

Merge–split samplers with sequential allocation for conjugate random-partition models (the R package `sams`). No convergence theory was seen.

D8 Particle Gibbs split-merge sampling for Bayesian inference in mixture models [K random] [Q1] FT

A. Bouchard-Côté, A. Doucet, A. Roth (2017). *J. Mach. Learn. Res. 18(28), 1–39; arXiv: 1508.02663*

Dirichlet-process mixtures, plus one example with $K = 5$ fixed. The theory covers invariance and irreducibility only.

D9 Sampling from Dirichlet process mixture models with unknown concentration parameter: mixing issues in large data implementations [K random] [context] Abs

D. I. Hastie, S. Liverani, S. Richardson (2015). *Statist. Comput. 25, 1023–1037*

An empirical study of mixing problems in Dirichlet-process mixtures, with no ergodicity or rate theory.

D10 Many processors, little time: MCMC for partitions via optimal transport couplings [K random] [Q2] FT

T. D. Nguyen, B. L. Trippe, T. Broderick (2022). *Proc. AISTATS, PMLR 151, 3483–3514; arXiv: 2202.11258*

Conjugate Gaussian DP mixtures and graph-colouring partitions. Single-site Gibbs and Jain–Neal split–merge chains are coupled by an optimal-transport coupling under a Hamming-type partition metric, plus a small independent "nugget". The coupled estimator is unbiased with finite variance and finite expected cost (Theorem 1, Corollary 1). The meeting time satisfies $\mathbb{P}(\tau > t) \le (1 - \epsilon^N)^t$, which degenerates as N grows, and the total-variation bound $C\alpha^t$ is not explicit. There are no mixing rates.

D11 Complexity bounds for Dirichlet process slice samplers [K random] [context] Abs

B. Franzolini, F. Gaffi (2026). *arXiv: 2602.00878 (v2, May 2026)*

High-probability bounds on the per-iteration cost of Dirichlet-process slice samplers, which is random. The extra overhead relative to the number of posterior clusters grows at most logarithmically in n, with no likelihood assumptions. This bounds cost, not mixing.

D12 Bayesian sparse Gaussian mixture model in high dimensions [K random] [context] FT (partial)

D. Yao, F. Xie, Y. Xu (2022). *arXiv: 2207.10301 (a search listing suggests JMLR vol. 26[†])*

Proves posterior contraction and mis-clustering rates for a sparse high-dimensional Gaussian mixture with a random number of clusters. Nothing is proved about its Gibbs sampler.

1.6 Stochastic block models and relatives with a random number of blocks

Summary. No paper was found that proves ergodicity beyond the automatic finite-state argument, or any quantitative bound, for an MCMC sampler on an SBM-type posterior with random K. Every random-K SBM sampler found is at level [Q0], or [Q1] with an informal argument.

- Fully collapsed samplers on partitions of $[n]$ (S9, S7, S13), and allocation samplers with $k \le n$ (S8), have a finite state space. They are therefore uniformly ergodic once irreducible and aperiodic.
- Samplers that keep continuous block parameters (S10, S14, S11, S15, S16), or allow an unbounded number of empty blocks (S3, S12), do not get this for free. Nobody has addressed them.

S1 Estimation and prediction for stochastic blockstructures [K known] [Q0] Abs Sec

K. Nowicki, T. A. B. Snijders (2001). *J. Amer. Statist. Assoc. 96(455), 1077–1087.* doi: 10.1198/016214501753208735[†]

The baseline Bayesian SBM. K is fixed by the user, and different values are compared through summary statistics. The sampler is Gibbs, with label switching handled by permutation-invariant summaries. The software manual says there are "no definite guidelines" for burn-in and suggests 10^4–10^5 iterations and repeated runs.

S2 Learning systems of concepts with an infinite relational model [K random] [Q0] FT

C. Kemp, J. B. Tenenbaum, T. L. Griffiths, T. Yamada, N. Ueda (2006). *Proc. AAAI-06, 381–388*

The infinite relational model: each type's partition has a CRP(γ) prior, and the Beta link parameters are integrated out. The paper says MCMC "can" be used (citing Jain–Neal), but the results come from hill-climbing to the MAP partition with move, split and merge proposals. There is no convergence statement.

S3 Improved Bayesian inference for the stochastic block model with application to large networks [K random] [Q1] FT

A. McDaid, T. B. Murphy, N. Friel, N. J. Hurley (2013). *Comput. Statist. Data Anal. 60, 12–31.* doi: 10.1016/j.csda.2012.10.021; *arXiv:* 1203.3083

$K \sim$ Poisson(1) conditioned on $K > 0$, with proportions and block parameters collapsed. The allocation sampler has four Metropolis–Hastings moves: add or remove an empty cluster, single-node Gibbs, M3, and absorb–eject. The paper calls it "an irreducible ergodic Markov chain algorithm". It says the add/remove and single-node moves "are sufficient to sample the space but have slow mixing". No proof is given, and the paper says "we have not defined a convergence criterion".

S4 Efficient Monte Carlo and greedy heuristic for the inference of stochastic block models [Q1] FT

T. P. Peixoto (2014). *Phys. Rev. E 89, 012804.* doi: 10.1103/PhysRevE.89.012804; *arXiv:* 1310.4378

The number of blocks B is fixed within a run and selected separately by description length. The sampler makes single-node Metropolis–Hastings moves with a neighbour-informed proposal. The proposal "fulfills the ergodicity condition, but not detailed balance", which is corrected by a Metropolis–Hastings step. Speed is shown only empirically.

S5 Nonparametric Bayesian inference of the microcanonical stochastic block model [K random] [Q1] FT

T. P. Peixoto (2017). *Phys. Rev. E 95, 012317.* doi: 10.1103/PhysRevE.95.012317; *arXiv:* 1610.02703

$P(B) = 1/N$ on $[1, N]$, with degree correction and nested priors. Single-node Metropolis–Hastings moves run at each level, including moves into a new group. "Any choice with nonzero probability for all values of s will preserve ergodicity". Equilibration time depends strongly on the proposals.

S6 Nonparametric weighted stochastic block models; Bayesian stochastic blockmodeling [K random] [Q1] FT

T. P. Peixoto (2018, 2019). *Phys. Rev. E 97, 012306 (arXiv:* 1708.01432*);* in Advances in Network Clustering and Blockmodeling, *Wiley (arXiv:* 1705.10225*)*

The same informal ergodicity arguments. The book chapter warns that "the asymptotic properties of the Markov chain may take a very long time to be realized".

S7 Merge-split Markov chain Monte Carlo for community detection [K random] [Q1] FT

T. P. Peixoto (2020). *Phys. Rev. E 102, 012305; arXiv:* 2003.07070

Random K in degree-corrected and nested SBMs, with parameters integrated out. Merge–split moves with exactly computed reverse probabilities, via auxiliary "staging" splits. Ergodicity is argued informally. The evidence is empirical:

- mixing is "several orders of magnitude" faster than with single-node moves;
- single-node moves are effectively non-ergodic on small networks (for the football network, the easiest single-node move has posterior ratio about 6×10^{-8});
- the paper's own figures show incomplete convergence on some networks.

S8 Estimating the number of communities in a network [K random] [Q0] FT

M. E. J. Newman, G. Reinert (2016). *Phys. Rev. Lett. 117, 078301.* doi: 10.1103/PhysRevLett.117.078301; *arXiv:* 1605.02753

Uniform prior on $k \in \{1, \dots, n\}$, with empty groups allowed, for Poisson SBMs and degree-corrected SBMs. The moves are heat-bath node moves and $k \pm 1$ moves. Neither ergodicity nor detailed balance is stated. The chain "can get stuck in metastable states", so restarts are used.

S9 Efficient method for estimating the number of communities in a network [K random] [Q1] FT

M. A. Riolo, G. T. Cantwell, G. Reinert, M. E. J. Newman (2017). *Phys. Rev. E 96, 032310.* doi: 10.1103/PhysRevE.96.032310; *arXiv:* 1706.02324

A restaurant-type prior on (g, k) with $P(k) \approx e^{-1}/(k-1)!$, for degree-corrected SBMs. Metropolis node moves include moving a node to a new group. Ergodicity "is trivial to show", and detailed balance is derived. Yet "the Monte Carlo algorithm appears not to equilibrate fully when the number of groups becomes large". It underestimates k, while runs started from the planted partition recover

it. This is the clearest documented instance of the phenomenon in P3.A–B.

S10 Probabilistic community detection with unknown number of communities [K random] [Q0] FT

J. Geng, A. Bhattacharya, D. Pati (2019). *J. Amer. Statist. Assoc. 114(526), 893–905. doi:10.1080/01621459.2018.1458618; arXiv:1602.08062*

The MFM-SBM: $k \sim$ truncated Poisson(1), $\pi \mid k \sim \text{Dir}(\gamma)$, $Q_{rs} \sim \text{Beta}(a, b)$. A Gibbs sampler integrates out k and π through the MFM urn and draws Q from its Beta conditionals. The statistical theory, for balanced homogeneous SBMs with $n\bar{D}(p_0, q_0)/K \to \infty$, is:

- Theorem 4.1, a Bayes-risk bound for known K;
- Corollary 4.2, posterior mass on the true clustering tending to one;
- Theorem 4.3, $\Pi(k = K \mid A) \geq 1 - \exp(-cn^q)$, for $K \in \{2, 3\}$ only.

General unknown K is "beyond the scope". Convergence of the sampler is shown only empirically.

S11 Inference for a generalised stochastic block model with unknown number of blocks and non-conjugate edge models [K random] [Q0] FT

M. Ludkin (2020). *Comput. Statist. Data Anal. 152, 107051†. doi:10.1016/j.csda.2020.107051; arXiv:1909.09421*

$K - 1 \sim \text{Poisson}(\delta)$, with weighted edges of arbitrary non-conjugate distributions. Reversible jump with split/merge, add/delete of empty blocks, reallocation and random-walk updates. Acceptance ratios are derived, but irreducibility and ergodicity are never stated: "assessing the convergence of a reversible jump Markov chain is non-trivial".

S12 Bayesian estimation of the latent dimension and communities in stochastic blockmodels [K random] [Q0] FT

F. Sanna Passino, N. A. Heard (2020). *Statist. Comput. 30(5), 1291–1307. doi:10.1007/s11222-020-09946-6; arXiv:1904.05333*

$K \sim \text{Geometric}(\omega)$ with $\text{Dir}(\alpha/K)$ weights, and a random latent dimension. Collapsed Gibbs plus split–merge, create/remove empty cluster, and dimension-changing moves. Convergence is checked only by running different chains.

S13 Extended stochastic block models with application to criminal networks [K random] [Q0] FT

S. Legramanti, T. Rigon, D. Durante, D. B. Dunson (2022). *Ann. Appl. Statist. 16(4), 2369–2395. doi:10.1214/21-AOAS1595; arXiv:2007.08569*

Gibbs-type partition priors (Dirichlet-multinomial, DP, Pitman–Yor, MFM, Gnedin) with collapsed Gibbs. Traceplots "suggest rapid convergence and effective mixing". Nothing is proved about the sampler.

S14 Consistent Bayesian community detection [K random] [Q0] FT

Z. Jiang, S. T. Tokdar (2021). *arXiv:2101.06531; no journal version found*

$K \sim \text{Poisson}(1)$, with a diagonally dominant (assortative) prior on block probabilities. Theorem 3.5 shows that the posterior probability of $\{K = k_0\} \cap \{Z = Z_0\}$ tends to one for $k_0 \lesssim \sqrt{n}$, with a sparse extension (Theorem 6.2). The moves are McDaid-type plus Beta draws. There is no MCMC theory, and simulations use $n \leq 75$.

S15 Bayesian nonparametric community detection in assortative stochastic block models [K random] [Q0] FT

M. Amongero, P. De Blasi (2025). *arXiv:2506.19576 (v1 titled* Bayesian community detection in assortative stochastic block model with unknown number of communities*)*

A Gnedin prior on k, with assortativity imposed through a random cutoff. Gibbs sampling with auxiliary blocks in the style of Neal's Algorithm 8. There are no theorems, and convergence is checked by $\hat{R} < 1.1$.

S16 Infinite-degree-corrected stochastic block model [K random] [Q0] FT

T. Herlau, M. N. Schmidt, M. Mørup (2014). *Phys. Rev. E 90(3), 032819. doi:10.1103/PhysRevE.90.032819; arXiv:1311.2520*

A CRP prior, with Gibbs on labels and random-walk updates of hyperparameters. Runs of 1000 iterations "may exhibit mixing times significantly longer".

S17 Block clustering with collapsed latent block models [K random] [Q0] FT

J. Wyse, N. Friel (2012). *Statist. Comput. (2012)†; arXiv:1011.2948*

Latent block model with truncated Poisson(1) priors on the numbers of row and column clusters. A collapsed allocation sampler, with only integrated-autocorrelation diagnostics.

S18 A Bayesian nonparametric stochastic block model for directed acyclic graphs [K random] [Q0] FT
C. Lee, M. Battiston (2023). *arXiv: 2301.07513 (a JCGS version†)*

A Pitman–Yor partition prior. The chain is "occasionally getting stuck at $K_n = 4$".

S19 Bayesian community detection for networks with covariates [K random] [Q0] FT
X. Shen, A. A. Amini, N. Josephs, L. Lin (2025). *Bayesian Anal. 20(3), 735–762. doi: 10.1214/24-BA1415; arXiv: 2203.02090*

A covariate-dependent product-partition prior with DP-type cohesion and a Gibbs sampler. "Good mixing" is claimed empirically only.

S20 Two further SBM samplers [Q0] Meta/FT
M. Mørup, M. N. Schmidt (2012); Zhao, Ye & Huang (2021) (2012, 2021). *Neural Comput. 24(9), 2434–2456; Entropy 23(1), 65 (doi: 10.3390/e23010065)*

- Mørup & Schmidt, *Bayesian community detection*: only the citation was confirmed.
- Zhao, Ye & Huang (title and initials not recorded): the Metropolis run length is chosen empirically, and it is "unknown whether $O(n \log n)$ is sufficient".

S21 Statistical theory for Bayesian SBMs (no MCMC) [context] Abs
S. L. van der Pas, A. W. van der Vaart (2018); J. van Waaij, B. J. K. Kleijn (2020); Ghosh, Pati, Bhattacharya (2020) (2018–2020). *Bayesian Anal. 13(3), 767–796; arXiv: 2005.01362; Sankhyā A 82(2), 448–476*

- *Bayesian community detection*: K known, strong consistency of the posterior mode when the expected degree $\gg (\log n)^2$.
- *Uncertainty quantification in the stochastic block model with an unknown number of classes*: contraction rates, and consistent posterior-odds tests for the number of classes.
- Ghosh, Pati & Bhattacharya (title not recorded): posterior contraction rates. How K is treated was not checked.

S22 Evidence thresholds of collapsed structured block models [context] Abs
Papamichalis, Ruane (2026). *arXiv: 2601.02828*

The summed evidence switches at the Kesten–Stigum threshold. The best single partition keeps the planted structure only above a higher, first-moment threshold. This matters for what a random-K sampler is actually targeting.

1.7 Quantitative results with K random: partial progress

Besides the decomposition in T13, the coupling bounds in D10 and the cost bounds in D11, the partial results are these.

R1 A fast non-reversible sampler for Bayesian (finite) mixture models [K known] [Q3] FT (main text)
F. Ascolani, G. Zanella (v1); F. Ascolani, P. Manildo, G. Zanella (v2) (2025, 2026). *arXiv: 2510.03226: v1 (3 Oct 2025), titled* . . . for Bayesian finite mixture models*; v2 (1 Oct 2026),* . . . for Bayesian mixture models

- *Samplers.* Four samplers on $[K]^n$ are compared: marginal (collapsed) Gibbs, the conditional sampler of Diebolt & Robert, a reversible Metropolis–Hastings move over pairs of clusters, and its lifted non-reversible version.
- *Variance comparison.* For lazy kernels and any target, the non-reversible sampler's asymptotic variance is at most that of the reversible one, which is at most 4 times that of marginal Gibbs, plus $3\mathrm{Var}_\pi(g)$ (v2, Theorem 1). Marginal Gibbs beats the conditional sampler for every likelihood (a Peskun-type result).
- *Scaling limits, prior only.* Marginal Gibbs, with time scaled by $n^2/2$, converges to a Wright–Fisher diffusion. The non-reversible sampler, with time scaled by n, converges to a piecewise-deterministic process. This gives the heuristic $O(n^2)$ versus $O(n)$. The eigenvalue used comes from P1.
- *Random K.* v1 says "we stick to the case of a fixed number of components. . . additional technical difficulties that we leave to future work". v2 applies the method to Pitman–Yor and Dirichlet-process partitions, with empirical results. Its Remark 1 says that "Theorem 4 in Section D extends (17) to general random partition models", meaning the variance comparison. That theorem was

not seen. The proved quantitative results are therefore for fixed K, and the scaling limits for fixed K and the prior only.

The supplement, including Theorem 4, was not read. Ascolani's slides of January 2026 list the speed-up for the posterior as open. *This is the most active competing line for P3.E.*

R2 Non-reversible jump algorithms for Bayesian nested model selection [Q3] FT

P. Gagnon, A. Doucet (2021). *arXiv:1911.01340 (journal version not checked)*

Non-reversible versions of reversible jump. Proposition 1 gives invariance. Corollary 1 shows the non-reversible asymptotic variance is at most that of reversible jump. In the supplement, idealised model-index dynamics have scaling limits: reversible jump gives a diffusion on time scale $O(K_{\max}^2)$, the non-reversible version a piecewise-deterministic process on time scale $O(K_{\max})$. Mixtures are cited as an application but not treated.

R3 Weak convergence and optimal tuning of the reversible jump algorithm [Q3] Abs

P. Gagnon, M. Bédard, A. Desgagné (2019). *Math. Comput. Simulation 161, 32–51; arXiv:1612.05653*

The first diffusion limit for reversible jump, in a stylised nested-model setting. It recovers the 0.234 acceptance rule.

R4 Informed reversible jump algorithms [Q2] FT

P. Gagnon (2021). *Electron. J. Statist. 15(2). doi:10.1214/21-EJS1877*

Reversible by construction, with ergodicity from Tierney. As $n \to \infty$, under Bernstein–von Mises, informed reversible jump converges weakly to an ideal sampler. The example is a regression model; mixtures are not treated.

1.8 Quantitative results with K known: upper bounds

U1 Mixing time of Metropolis–Hastings for Bayesian community detection [K known] [Q3] FT

B. Zhuo, C. Gao (2021). *J. Mach. Learn. Res. 22(10), 1–89; arXiv:1811.02612*

The only rapid-mixing theorem for a Bayesian clustering posterior that is used in practice.

- *Model and prior.* A homogeneous SBM (within-block probability p, between-block q, $p > q$) with K a known constant. The true block sizes lie in $[n/(\beta K), \beta n/K]$. The prior is uniform on labellings in S_α (sizes in $[n/(\alpha K), \alpha n/K]$, $\alpha > \beta$), with Beta priors on the edge probabilities. The target is a tempered posterior Π^ξ, $\xi \geq 1$.
- *Sampler.* Single-node relabelling Metropolis–Hastings restricted to S_α, made lazy in the proofs. The authors say the restriction "seems to be necessary for the rapidly mixing", based on experiments.
- *Conditions.* The signal is above the exact-recovery threshold: $\liminf \bar{n} I / \log n > 1$, where $I \approx (\sqrt{p} - \sqrt{q})^2$ is a Rényi divergence and $\bar{n} = n/(K\beta)$. The start is warm, with misclassification at most γ_0; for $K \geq 3$ this needs $\gamma_0 = o(1)$, e.g. spectral clustering. The temperature satisfies a lower bound in terms of the signal margin.
- *Result.* With high probability, $\tau_\epsilon(Z_0) \leq 4Kn^2 \max\{\gamma_0, n^{-\tau}\} (\xi \log \Pi(Z_0 \mid A)^{-1} + \log \epsilon^{-1}) \lesssim n^2(n^2 I + \log \epsilon^{-1})$.
- *Method.* Posterior-ratio bounds plus canonical paths for the spectral gap. There are no lower bounds.

One report notes that the $\xi = 1$ corollary looks inconsistent with the paper's stated ξ thresholds, so it should be checked against the proof. Project P3.C(C3) extends this paper to random K.

U2 Sampling for Bayesian mixture models: MCMC with polynomial-time mixing [K known] [Q3] FT

W. Mou, N. Ho, M. J. Wainwright, P. L. Bartlett, M. I. Jordan (2019). *arXiv:1912.05153 (v1 only)*

Target: the power posterior of $\frac{1}{2}N(\theta, I_d) + \frac{1}{2}N(-\theta, I_d)$ with a flat prior, so $K = 2$ and the weights are known. The sampler is a random walk with a random sign flip. If $n \gtrsim d(\beta^2(1 + \|\theta_0\|^2) + d)\log(\cdot)$, the mixing time is polynomial, $T(\epsilon) \lesssim d^{3/2}(d + \|\theta_0\|^2)^{9/2} \log^{11/2}(1/\epsilon)$. No separation is needed. The tool is a way to combine conditional and marginal Poincaré inequalities. Left open: $K > 2$, general mixtures, and the untempered posterior.

U3 Sampling approximately low-rank Ising models: MCMC meets variational methods [K known] [Q3] FT

F. Koehler, H. Lee, A. Risteski (2022). *Proc. COLT, PMLR 178; arXiv:2202.08907*

Two clusters. Integrating out the mean direction turns the label posterior into a low-rank Ising model.

The paper also covers a contextual SBM. The sampler is simulated tempering on an extended space with Glauber dynamics inside, plus importance steps, so it is not a pure Markov chain. For the Gaussian mixture it runs in time polynomial in n for fixed p. The authors note that plain Glauber dynamics should not mix, because swapping the labels gives a bimodal posterior.

U4 Fast conditional mixing of MCMC algorithms for non-log-concave distributions [K known] [Q3] FT

X. Cheng, B. Wang, J. Zhang, Y. Zhu (2023). *NeurIPS 2023; arXiv: 2306.10506*

Langevin Monte Carlo for the power posterior of a symmetric two-component location mixture. The averaged iterates are close to the conditional posterior within each label-switching half-space. No mixing across the two modes is claimed.

U5 Sampling from constrained Gibbs measures: with applications to high-dimensional Bayesian inference [K known] [Q3] Abs (partial)

R. Wang, X. Chen, S. Chewi (2026). *arXiv: 2602.22369*

Reflected Langevin dynamics has an essentially dimension-free Poincaré constant $O(1/n)$ on a "good set" around a mode on the constraint boundary, which gives convergence from a warm start. Gaussian mixtures are said to be covered when $n \gg d^2$. That section could not be read.

U6 Locally stationary distributions: a framework for analyzing slow-mixing Markov chains [K known] [Q3] FT

K. Liu, S. Mohanty, P. Raghavendra, A. Rajaraman, D. X. Wu (2024). *arXiv: 2405.20849 (v4, July 2025; FOCS 2024 according to an author's homepage†)*

Glauber dynamics for an Ising model with coupling $(\beta/\sqrt{d})$ times the centred adjacency matrix of a sparse two-community SBM, i.e. a mismatched-temperature surrogate of the posterior, started anywhere. A constant factor above the Kesten–Stigum threshold, the chain stopped at a uniform random time in $[0, \tilde{\Theta}(n^{5+o(1)})]$ achieves weak recovery. The general theorem is that any reversible chain is locally stationary at a random time. The paper says Glauber dynamics on the true posterior "takes exponential time to mix… from a worst-case initialization", but this is motivation, not a result. Sampling the posterior is posed as open.

U7 Fast mixing in sparse random Ising models [K known] [Q3] Abs

K. Liu, S. Mohanty, A. Rajaraman, D. X. Wu (2024). *arXiv: 2405.06616 (FOCS 2024 according to an author's homepage†)*

Glauber dynamics mixes in $n^{1+o(1)}$ steps for the Viana–Bray model at high temperature, via a modified log-Sobolev inequality. The result extends to two-community SBM graphs.

U8 Markov chains approximate message passing [K known] [Q3] Abs

A. Rajaraman, D. X. Wu (2025). *arXiv: 2512.02384 (v3, July 2026)*

Glauber dynamics on the annealed posterior of a spiked Wigner model with a Boolean spike, the dense analogue of the two-community SBM. From a warm start it reaches the Bayes-optimal fixed point. The full phase transition is conditional on conjectured mixing of the SK model.

U9 Mixing of Glauber dynamics on high overlap Gibbs measures [K known] [Q3] FT

A. S. Bandeira, A. El Alaoui, A. Rödder (2026). *arXiv: 2607.06813*

Polynomial-time mixing via stochastic localisation and a spectral-gap bound. Corollary 4.1, whose proof is only sketched, gives polynomial mixing for the $\mathbb{Z}_2$-synchronisation posterior with spin-level side information. The case without side information is open.

U10 Rapidly mixing multiple-try Metropolis algorithms for model selection problems [Q3] FT

H. Chang, C. Lee, Z. T. Luo, H. Sang, Q. Zhou (2022). *NeurIPS 2022; arXiv: 2207.00689*

Under a unimodality condition, multiple-try Metropolis with locally balanced weights improves path-method mixing bounds by the number of tries. The SBM appears only in simulations. The paper says the Zhuo–Gao posterior satisfies the condition but proves no SBM theorem.

U11 The Metropolis algorithm for graph bisection [K known] [Q3] Abs

M. Jerrum, G. B. Sorkin (1998). *Discrete Appl. Math. 82†. doi: 10.1016/S0166-218X(97)00133-9*

In a planted bisection model with a large enough gap, fixed-temperature Metropolis finds the planted bisection in $O(n^{2+\epsilon})$ steps with high probability. This is a hitting-time result, not a mixing result.

U12 Semi-supervised community detection using Glauber dynamics for an Ising model; Posterior sampling in high dimension via diffusion processes [K known] [context] Abs/FT

K. Avrachenkov, D. Goldsztajn (2025); A. Montanari, Y. Wu (2023) (2023, 2025). *arXiv: 2506.09223; arXiv: 2304.11449*

The first gives almost exact recovery for a two-community SBM with side information in quasi-linear time, which is not a mixing result. The second is a non-MCMC benchmark: a diffusion and approximate message passing sampler for spiked models with discrete priors.

1.9 Quantitative results with K known: lower bounds and slow mixing

L1 Lower bounds for the Gibbs sampler over mixtures of Gaussians [K known] [Q3] FT

C. Tosh, S. Dasgupta (2014). *Proc. ICML, PMLR 32(2), 1467–1475*

Spherical Gaussian mixture with known σ^2, centres from $N(\mu_0, \sigma_0^2 I_d)$ and Dir(α) weights, with k fixed. The sampler is random-scan collapsed Gibbs on labels, analysed as a chain on partitions. The tool is conductance, $\tau \geq 1/(4\Phi)$.

- *Misspecified k* (Theorem 6.1). Six tight clusters at the vertices of a prism, fitted with $k = 3$, give $\tau_{\text{mix}} \geq \frac{1}{24}\exp(r^2/(8\sigma^2))$. Better separation means slower mixing.
- *Correct* $k = 3$ (Theorem 6.2). Clusters at the corners of a triangle give a bound that is the minimum of an exponential in r^2/σ^2 and a polynomial in n, summarised as "at least $n^{\Omega(\alpha)}$ steps". The exact exponent was not recovered.

There are no upper bounds. This is the main template for the lower bounds in P3.A. There, the bottleneck comes from dimension and from the move that changes K, not from separation alone.

L2 Sampling from the greedy mixture posterior [K known] [Q3] Sec (a summary of the full text)

D. P. O'Connell (2021). *PhD dissertation, Yale University*

Two components: one Gaussian with unknown centre (conjugate prior), the other a fixed density. Collapsed Gibbs, plus tempering variants.

- Theorem 2.2.1 (conductance): the mixing time is at least the minimum of two exponentials, one in the cluster isolation and one in the centre separation.
- Theorem 3.4.4: the bottleneck can persist under internal-annealing tempering.
- Theorem 4.3.2: under subsample annealing, removing one datum can suffice to keep the mixing slow.

The theorem statements were not checked line by line.

L3 Convergence rate of Markov chain methods for genomic motif discovery [K known] [Q3] FT

D. B. Woodard, J. S. Rosenthal (2013). *Ann. Statist. 41(1), 91–124. doi: 10.1214/12-AOS1075*

A two-component (motif or background) mixture with Dirichlet parameters integrated out, sampled by random-scan Gibbs. If the data contain more than one true motif, the spectral gap decays exponentially in the sequence length (Theorem 3.2). For motif width one it decays only polynomially (Theorem 3.3).

L4 The stochastic block model has the overlap graph [sic] property for modularity [K known] [Q3] FT

S. Bhamidi, D. Gamarnik, R. van der Hofstad, N. Litvak, P. Prałat, F. Skerman, M. Tousinejad (2026). *arXiv: 2605.10911*

The arXiv title reads "graph", while the text says "gap". SBM with $k \geq 3$ equal blocks, $p = \omega a/n$ and $q = \omega b/n$ with ω above a constant. Near-optimal-modularity partitions are either close to the planted partition or far from it (Theorem 2.2, an overlap gap). Single-vertex heat-bath dynamics for $\exp(\beta n \cdot \text{modularity})$ at large β, started from the far set, need at least e^{cn} steps (Theorem B.2). This is a modularity-tilted measure, not the Bayesian posterior.

1.10 Chains on partitions without data

These are prior-only analogues: the dynamics of a sampler when the likelihood is flat.

P1 Rates of convergence of some multivariate Markov chains with polynomial eigenfunctions [no data] [Q3] FT

K. Khare, H. Zhou (2009). *Ann. Appl. Probab. 19(2), 737–777. doi: 10.1214/08-AAP562*

Exact spectra for chains with Dirichlet-multinomial stationary laws: Moran, Hubbell's "local community" chain and the Dirichlet-multinomial Gibbs sampler. The local-community chain has $\beta_1 = 1 - |\alpha|/(N(N + |\alpha| - 1))$, so it needs about N^2 steps. Mixtures are never mentioned. Ascolani–Zanella (R1) identify this eigenvalue with prior-case collapsed Gibbs on allocations for fixed K.

P2 Convergence time to the Ewens sampling formula in the infinite alleles Moran model [no data] [Q3] Abs

J. C. Watkins (2010). *J. Math. Biol. 60(2), 189–206. doi: 10.1007/s00285-009-0255-x*

The discrete-time infinite-alleles Moran model, with population size M and mutation probability μ, has a cutoff at about $\log(M\mu)/\mu$ generations. Its stationary law under sampling without replacement is the Ewens sampling formula, and the total-variation profile is Gumbel. *Our identification:* with no data, random-scan single-site collapsed Gibbs for a CRP(α) partition removes a point and reinserts it in block c with probability $n_c/(n-1+\alpha)$, or in a new block with probability $\alpha/(n-1+\alpha)$. This is the infinite-alleles Moran model with $\mu = \alpha/(n-1+\alpha)$, if the parent is drawn from the other $n-1$ individuals. Here $M\mu \to \alpha$, so Watkins' cutoff is informative only when α grows with n. For fixed α the dynamics are diffusive, on a time scale of order n^2 updates. The identification also needs a random scan. The prior-only, random-K case therefore reduces to population-genetics models, and the open questions concern the posterior.

P3 Gibbs sampling, exponential families and orthogonal polynomials [no data] [Q3] FT

P. Diaconis, K. Khare, L. Saloff-Coste (2008). *Statist. Sci. 23(2), 151–178. doi: 10.1214/07-STS252*

Sharp rates for two-component Gibbs samplers in conjugate exponential and location families. Mixtures are not covered.

P4 Mixing time of Metropolis chain based on random transposition walk converging to multivariate Ewens distribution [no data] [Q3] FT

Y. Jiang (2015). *Ann. Appl. Probab. 25(3), 1581–1615. doi: 10.1214/14-AAP1031*

Total-variation cutoff at $\frac{1}{2}\max(1, 1/\theta)\, n \log n$, started from the identity. The cycle type performs a split–merge chain on integer partitions. In the paper's notation θ is the inverse of the Ewens parameter.

P5 Uniform split–merge and coagulation–fragmentation chains [no data] [Q2] Abs

P. Diaconis, E. Mayer-Wolf, O. Zeitouni, M. P. W. Zerner (2004); Mayer-Wolf, Zeitouni, Zerner (2001); O. Schramm (2005) (2001–2005). *Ann. Probab. 32(1B), 915–938; arXiv: math/0105111; Israel J. Math. 147, 221–243*

Poisson–Dirichlet(1) is the unique invariant law of uniform split–merge on interval partitions (Vershik's conjecture). After cn random transpositions with $c > 1/2$, rescaled cycle sizes converge to PD(1). There are no rates.

P6 Mixing times for random k-cycles and coalescence-fragmentation chains [no data] [Q3] Abs

N. Berestycki, O. Schramm, O. Zeitouni (2011). *Ann. Probab. 39(5), 1815–1843. doi: 10.1214/10-AOP634*

Random k-cycles have cutoff at $(1/k)\, n \log n$. The proof analyses the cycle structure as a coalescence–fragmentation chain.

P7 A probabilistic interpretation of the Macdonald polynomials [no data] [Q3] Abs

P. Diaconis, A. Ram (2012). *Ann. Probab. 40(5)†, 1861–1896. doi: 10.1214/11-AOP674*

An auxiliary-variable chain on partitions whose stationary law is a Macdonald measure, with the Ewens law as a limit. The eigenvalues are explicit.

P8 Convergence rates of Markov chains on spaces of partitions [no data] [Q3] FT

H. Crane, S. P. Lalley (2013). *Electron. J. Probab. 18(61), 1–23. doi: 10.1214/EJP.v18-2389*

Exchangeable "cut-and-paste" chains mix in $O(\log n)$ steps, with cutoff under a density assumption. Gibbs and split–merge chains are not covered.

P9 Mixing times of a Burnside process Markov chain on set partitions [no data] [Q3] FT

J. E. Paguyo (2022). *arXiv: 2207.14269 (Adv. Appl. Math., 2026†)*

Uniform set partitions with at most k blocks. $t_{\text{mix}} \le \lceil 2k \log(n/\epsilon) \rceil$ for $k \ge n$, and $\le \lceil (k-1)! \log(1/\epsilon) \rceil$ for $k < n$.

P10 Up-down ordered Chinese restaurant processes with two-sided immigration, emigration and diffusion limits [no data] [Q3] Abs

Q. Shi, M. Winkel (2020). *arXiv: 2012.15758 (v2, December 2025)*

Scaling limits of up-down ordered Chinese restaurant chains, which are prior-only analogues of partition Gibbs samplers, to interval-partition diffusions.

P11 A reversible allelic partition process and Pitman sampling formula [no data] [Q2] Abs

M. Giordano, P. De Blasi, M. Ruggiero (2019). *arXiv: 1905.05154*

A continuous-time birth, death and immigration chain on allelic partitions. It is reversible, with a negative-binomial mixture of Pitman sampling formulae as stationary law. There are no rates.

1.11 Trans-dimensional and discrete-space templates outside clustering

V1 On the computational complexity of high-dimensional Bayesian variable selection [Q3] FT

Y. Yang, M. J. Wainwright, M. I. Jordan (2016). *Ann. Statist. 44(6), 2497–2532. doi: 10.1214/15-AOS1417*

Random-walk Metropolis–Hastings with add, delete and swap moves for a g-prior with model size truncated at s_0. From the worst start, $\tau_\epsilon \lesssim ps_0^2(\alpha(n+s_0)\log p + \log(1/\epsilon))$ with high probability (Theorem 2). The method is canonical paths plus posterior concentration. This is the closest template for a trans-dimensional upper bound, and the proof route proposed for P3.A.

V2 Dimension-free mixing for high-dimensional Bayesian variable selection [Q3] FT

Q. Zhou, J. Yang, D. Vats, G. O. Roberts, J. S. Rosenthal (2022). *J. R. Stat. Soc. Ser. B 84(5), 1751–1784. doi: 10.1111/rssb.12546*

Locally informed, thresholded Metropolis–Hastings mixes in a number of steps independent of p, from any start, under a unimodality condition.

V3 Complexity analysis of Bayesian learning of high-dimensional DAG models and their equivalence classes [Q3] FT

Q. Zhou, H. Chang (2023). *Ann. Statist. 51(3), 1058–1085. doi: 10.1214/23-AOS2280*

A general bound for finite spaces. Suppose neighbourhoods have size at most p^{t_1}, every state other than the mode has a neighbour with posterior ratio at least p^{t_2}, and $t_2 > t_1$. Then $T_{\text{mix}} \leq 2C\log(4/\pi_{\min})/\min_\theta P(\theta, N^*(\theta))$. This "unimodality through local moves" criterion is the natural tool for the split–merge upper bound in P3.A.

V4 Dimension-free relaxation times of informed MCMC samplers on discrete spaces [Q3] FT/Abs

H. Chang, Q. Zhou (2024). *arXiv: 2404.03867*

Relaxation time $O(\log 1/\pi_{\min})$ for informed samplers under unimodality. Multimodal targets are handled with restricted gaps and a warm start.

V5 Dimension-free mixing times of Gibbs samplers for Bayesian hierarchical models [Q3] Abs Sec

F. Ascolani, G. Zanella (2024). *Ann. Statist. 52, 869–894*

Mixing bounds for Gibbs samplers derived from Bayesian asymptotics. The template is that posterior concentration implies fast mixing.

V6 On mixing rates for Bayesian CART [Q3] Abs

J. Kim, V. Ročková (2025). *Electron. J. Statist. doi: 10.1214/25-EJS2397; arXiv: 2306.00126*

Polynomial mixing under hierarchical connectivity of the signal. Grow–prune moves cannot reach deep, isolated signals in better than superpolynomial time, while "twiggy" moves restore polynomial mixing. This is the closest analogue of "the move set decides polynomial versus exponential" in P3.A.

V7 Mixing-time lower bounds for BART [Q3] Abs

O. Ronen, T. Saarinen, Y. S. Tan, J. Duncan, B. Yu (2022); Y. S. Tan, O. Ronen, T. Saarinen, B. Yu (2024) (2022, 2024). *arXiv: 2210.09352; arXiv: 2406.19958*

For simplified single-tree BART, the lower bound grows exponentially in n. The time to hit a high-posterior-density set increases with n.

V8 Reversible-jump ergodicity proofs outside mixtures [Q2] FT/Sec

C. Andrieu, A. Doucet (1999); J. Vermaak, C. Andrieu, A. Doucet, S. J. Godsill (2004); Ortner et al. (2006); Schreck et al. (1999–2016). *IEEE Trans. Signal Process. 47(10), 2667–2676; J. Time Ser. Anal. 25(6), 785–809; in MCQMC 2004, 389–401; IEEE J. Sel. Top. Signal Process. 10(2), 366–375*

Uniform geometric ergodicity of reversible jump for sinusoid detection, via a Doeblin condition with a truncated Poisson prior on k and bounded frequencies. Geometric convergence is claimed for autoregressive model selection, with the proof in a technical report. Qin (T13) cites the last two as further geometric-ergodicity results. Andrieu & Doucet rely on a truncated prior and bounded frequencies. The conditions of the others were not checked.

1.12 Tempering

E1 Conditions for rapid mixing of parallel and simulated tempering on multimodal distributions [Q3] FT

D. B. Woodard, S. C. Schmidler, M. Huber (2009). *Ann. Appl. Probab. 19(2), 617–640. doi: 10.1214/08-AAP555*

Lower bounds on the gap of parallel and simulated tempering in terms of overlap between temperatures and within-mode gaps. Examples include mean-field Ising and two-component normal mixtures in $\mathbb{R}^M$. These are mixture-*shaped* targets, not mixture-model posteriors.

E2 Sufficient conditions for torpid mixing of parallel and simulated tempering [Q3] FT

D. B. Woodard, S. C. Schmidler, M. Huber (2009). *Electron. J. Probab. 14(29), 780–804*

Upper bounds on the gap through conductance, "persistence" and overlap. Torpid examples are unequal-variance normal mixtures, mean-field Potts with $q \geq 3$, and mean-field Ising with fixed temperatures. Mixture posteriors are not treated. This is the template for P3.F.

E3 Beyond log-concavity: provable guarantees for sampling multi-modal distributions using simulated tempering Langevin Monte Carlo [Q3] FT (partial)

R. Ge, H. Lee, A. Risteski (2017). *arXiv: 1710.02736 (venue not checked)*

Polynomial-time sampling for mixtures of Gaussians with a common spherical covariance. The motivation is the posterior of the means of a Gaussian mixture, but the theorem is for mixture-shaped targets.

E4 Restricted spectral gap decomposition for simulated tempering targeting mixture distributions [Q3] Sec

Garg, Balasubramanian, Zhou (2025). *NeurIPS 2025; arXiv: 2505.15059*

A decomposition theorem for restricted spectral gaps. For Gaussian-mixture targets, simulated tempering with random-walk Metropolis–Hastings needs a number of steps polynomial in the separation and logarithmic in the accuracy, but exponential in the dimension.

1.13 Lower-bound templates from planted problems

H1 Almost-linear planted cliques elude the Metropolis process [Q3] Abs

Z. Chen, E. Mossel, I. Zadik (2023). *SODA 2023†; Random Struct. Alg. (2025)†, doi: 10.1002/rsa.21274; arXiv: 2204.01911*

The Metropolis process fails to find planted cliques of size n^α for any $\alpha < 1$, from the empty clique and over a wide range of temperatures. Simulated tempering also fails.

H2 The landscape of the planted clique problem: dense subgraphs and the overlap gap property [Q3] Abs

D. Gamarnik, I. Zadik (2024). *Ann. Appl. Probab. 34(4), 3375–3434. doi: 10.1214/23-AAP2003*

The overlap gap property is proved for small polynomial clique sizes. A conditional MCMC lower bound in the arXiv version may not have been kept in the journal version.

H3 Free energy wells and overlap gap property in sparse PCA [Q3] Abs

G. Ben Arous, A. S. Wein, I. Zadik (2020). *Proc. COLT, PMLR 125, 479–482; arXiv: 2006.10689*

Free-energy wells imply exponential hitting times for a class of local MCMC from a worst-case start, at any temperature. This is a template for the free-energy argument in P3.A.

H4 On the low-temperature MCMC threshold: the cases of sparse tensor PCA, sparse regression, and a geometric rule [Q3] FT

Z. Chen, C. Sheehan, I. Zadik (2024). *arXiv: 2408.00746*

A checkable "spanning tree of energy-increasing paths" condition gives polynomial-time MAP finding. On the negative side, the overlap gap property gives super-polynomial lower bounds for reversible local chains.

H5 Almost-optimal local-search methods for sparse tensor PCA [Q3] FT

Lovig, Sheehan, Tsirkas, Zadik (2025). *arXiv: 2506.09959*

Changing the move set plus a warm start reaches the conjectured algorithmic threshold, bypassing the previous barrier. A move-set analogue of split–merge.

H6 Finding planted cliques using gradient descent [Q3] FT

R. Gheissari, A. Jagannath, Y. Xu (2023). *arXiv: 2311.07540 (v1: ...* using Markov chain Monte Carlo*); SIAM J. Discrete Math.†, doi: 10.1137/24M1680489*

Started from the full vertex set, a low-temperature local chain recovers cliques of size $\geq C\sqrt{n}$ in $O(n)$ steps. From the empty set it gets stuck. This is the planted-clique version of the asymmetry between initialisations in P3.A.

H7 Further hardness tools for local samplers [Q3] Abs/FT

Bandeira, El Alaoui, Hopkins, Schramm, Wein, Zadik (2022); Bandeira, Maillard, Nickl, Wang (2023); Chen, Mikulincer, Reichman, Wein (2025); Kunisky; Huang, Perkins, Potechin (2024) (2022–2025). *NeurIPS 2022; Phil. Trans. R. Soc. A 381(2247), 20220150; APPROX/RANDOM 2025 (LIPIcs 353, art. 47); arXiv: 2307.12581; arXiv: 2409.03974*

- The Franz–Parisi criterion is equivalent to low-degree hardness for Gaussian additive models, and implies failure of local MCMC.
- Free-energy barriers in Gaussian priors make cold-start MCMC fail for high-dimensional unimodal posteriors, with hitting times e^{cN}.
- Time lower bounds for the Metropolis process and simulated annealing.
- Optimality of Glauber dynamics for general-purpose Ising sampling.
- Hardness of sampling, for *stable* samplers, for the anti-ferromagnetic Ising model on random graphs.

H8 Asymptotic analysis of the stochastic block model for modular networks and its algorithmic applications [context] FT

A. Decelle, F. Krzakala, C. Moore, L. Zdeborová (2011). *Phys. Rev. E 84, 066106; arXiv: 1109.3041*

Cavity-method predictions, backed by numerics. Gibbs sampling equilibrates in linear time where belief propagation works. In the hard phases "the equilibration time of MCMC is exponentially large". Not rigorous. Angelini & Ricci-Tersenghi (Phys. Rev. X 13, 021011, 2023†; arXiv: 2206.04760) give numerical thresholds for simulated annealing on planted colouring.

H9 Low-degree lower bounds for clustering in moderate dimension [context] FT

A. Carpentier, N. Verzelen (2026). *arXiv: 2602.23023*

A low-degree lower bound for partially clustering isotropic Gaussian mixtures with $d \geq K$, plus a matching non-spectral algorithm. This is not about MCMC, but it gives the computational benchmark against which the posterior thresholds of P3.A should be compared.

1.14 Statistical context: what the samplers target

The journal entries C1, C3, C4, C5 and parts of C8 were checked for the companion bibliography; they are not in the four reports.

C1 Inconsistency of Pitman–Yor process mixtures for the number of components [context] Abs

J. W. Miller, M. T. Harrison (2014). *J. Mach. Learn. Res. 15, 3333–3370*

DP and PY mixtures are inconsistent for the number of components of a finite mixture. So with DP priors, "the right K" is the number of large clusters, not K_+.

C2 Consistency of mixture models with a prior on the number of components [context] Meta

J. W. Miller (2023). *Dependence Modeling 11(1)†. doi: 10.1515/demo-2022-0150; arXiv: 2205.03384*

Posterior consistency of the number of components under MFM priors. This is the fixed-dimension statistical counterpart of target A of P3.A.

C3 Asymptotic behaviour of the posterior distribution in overfitted mixture models [context] Abs

J. Rousseau, K. Mengersen (2011). *J. R. Stat. Soc. Ser. B 73(5), 689–710*

In overfitted mixtures with small Dirichlet weights (below $d/2$), the posterior empties the superfluous components. Opening an empty component is therefore the relevant move even when K is fixed, which links sparse finite mixtures to P3.A.

C4 Clustering consistency with Dirichlet process mixtures [context] Abs

F. Ascolani, A. Lijoi, G. Rebaudo, G. Zanella (2023). *Biometrika 110(2), 551–558*

With a prior on the concentration parameter, the DP-mixture posterior on the number of clusters

becomes consistent.

C5 Consistent model-based clustering using the quasi-Bernoulli stick-breaking process [context] Abs
C. Zeng, J. W. Miller, L. L. Duan (2023). *J. Mach. Learn. Res. 24(153)*
A stick-breaking prior with a consistent posterior on the number of clusters.

C6 Finite mixture models do not reliably learn the number of components [context] Meta
D. Cai, T. Campbell, T. Broderick (2021). *Proc. ICML 2021*
Under misspecification, the posterior number of components diverges. The samplers' target is fragile.

C7 Escaping the curse of dimensionality in Bayesian model-based clustering [context] Abs
N. K. Chandra, A. Canale, D. B. Dunson (2023). *J. Mach. Learn. Res. 24(144), 1–42; arXiv: 2006.02700*
With n fixed and $p \to \infty$, the posterior on partitions can degenerate either to one cluster or to all singletons, under conditions on the kernel and prior and whatever the true model. The setting differs from P3.A, which has known isotropic noise and growing n, where the posterior does concentrate. It shows how delicate the high-dimensional posterior is.

C8 Sparse finite mixtures, non-local and repulsive priors [context] Meta / Abs
G. Malsiner-Walli, S. Frühwirth-Schnatter, B. Grün (2016); J. Fúquene, M. Steel, D. Rossell (2019); F. Xie, Y. Xu (2020) (2016–2020). *Statist. Comput. 26†; J. R. Stat. Soc. Ser. B 81(5), 809–837; J. Amer. Statist. Assoc. 115(529), 187–203*
Sparse finite Gaussian mixtures with small Dirichlet weights empty superfluous components. Non-local priors relate Bayes factors for K to empty-cluster probabilities. Repulsive priors shrink the tail of the posterior on K. All of these change the prior's tail and its empty-component behaviour, which is what P3.D shows to matter for convergence.

1.15 What could not be verified

- *Primary text not read.* Tierney (1994), Green (1995), Preston (1976/77), Geyer & Møller (1994), Jain & Neal (2004), Nobile & Fearnside (2007) and Roberts & Rosenthal (2001). In particular, whether Geyer & Møller prove geometric ergodicity, and under which conditions, was not confirmed.
- *Technical details.*
  - The exact powers in Qin's Theorem 7.
  - The statement of Theorem 4 in Ascolani–Manildo–Zanella v2.
  - The polynomial exponent in Tosh & Dasgupta's Theorem 6.2.
  - The $\xi = 1$ corollary of Zhuo & Gao.
  - The uniform-ergodicity proof in Mukhopadhyay & Bhattacharya's supplement.
  - The conditions of Chandra, Canale & Dunson.
  - Section 5.2 of Wang, Chen & Chewi.
- *Bibliographic details marked †.* Pages and volumes from memory or a single secondary source, and conference venues taken from homepages.
- *Search gaps.* Planned queries left unrun when the budget ran out:
  - population reversible jump (Jasra, Stephens & Holmes 2007);
  - ergodicity of blocked Gibbs samplers (Ishwaran & James);
  - Sisson (2005);
  - Glauber slow mixing for SBM posteriors below the Kesten–Stigum threshold;
  - 2025–26 preprints on mixing of random-K samplers.

  Two targeted searches made for this document found no quantitative split–merge or high-dimensional DPM result. One was on "split–merge mixing time" and the other on "collapsed Gibbs high-dimensional new cluster".

2 Specific versions of P3: candidate projects

2.1 Overview

The literature leaves one question wide open. When, and why, are the moves that change K slow? The six projects below make this question concrete. P3.A is the flagship. P3.B transfers its mechanism to multilayer networks and so connects to P4. P3.C transfers it to ordinary SBMs, where the number of communities plays the role of the dimension. P3.D–P3.F are lower-risk or complementary. Table 3 summarises them.

	Question	Main target	Novelty	Risk	Venue
P3.A	Single-site Gibbs versus split–merge in high-dimensional Gaussian mixtures with a prior on K	e^{cp} lower bound for single-site moves, polynomial split–merge, posterior concentration	high	medium	AoS, JRSS-B
P3.B	The same in sparse multilayer SBMs, with layers in the role of dimension	$e^{cL \log n}$ lower bound; split proposals from P4's statistic	high	med.–high	AoS, JASA
P3.C	Why single-site samplers underestimate the number of communities in SBMs	$n^{\Omega(K)}$ barrier, underestimation law, posterior concentration with growing K, local mixing	high	medium	AoS, JASA
P3.D	Which random-K samplers are geometrically ergodic?	the tail of p_K decides; duality for exact-K samplers	medium	low	Biometrika, Bernoulli
P3.E	Posterior scaling limits in low dimension	Moran dynamics with selection; n^2 versus n	medium	medium (competition)	AoS, AAP
P3.F	Does tempering remove the barrier?	torpid tempering in the window of P3.A	medium	medium	AAP, AoS

Table 3: The six projects. "Novelty" is relative to Section 1.

2.2 Common framework

Collapsed posterior on partitions. Let C be a partition of $[n] = \{1, \dots, n\}$ with $t = |C|$ blocks. Write $n_c = |c|$ for block sizes, x_c for the data in block c, and $m(x_c) = \int \prod_{i \in c} f(x_i \mid \theta)\, G_0(d\theta)$ for the collapsed marginal likelihood. Two partition priors are used.

- *MFM* (M8): $K \sim p_K$ and $\mathrm{Dir}_K(\gamma)$ weights give

$$p(C) = V_n(t) \prod_{c \in C} \gamma^{(n_c)}, \qquad V_n(t) = \sum_k \frac{k_{(t)}}{(\gamma k)^{(n)}}\, p_K(k),$$

  where $x^{(m)}$ is the rising factorial and $k_{(t)}$ the falling factorial.

- *DP(α)*: $p(C) = \alpha^t \prod_c (n_c - 1)! / \alpha^{(n)}$.

The posterior is $\Pi_n(C) \propto p(C) \prod_{c \in C} m(x_c)$. It lives on a finite set, so every sampler below is uniformly ergodic (Route 1 in Section 1.2), and only rates are at stake.

Samplers.

(S1) *Single-site collapsed Gibbs*, as in Neal's Algorithm 3 or the MFM version. Point i is removed and reassigned, either to block c with weight $w_c\, m(x_i \mid x_c)$ or to a new block with weight $w_{\text{new}}\, m(x_i)$. The weights are $w_c = n_c$ and $w_{\text{new}} = \alpha$ for the DP, and $w_c = n_c + \gamma$ and $w_{\text{new}} = \gamma V_n(t+1)/V_n(t)$ for the MFM. The number of blocks changes only when a block is created from a single point or emptied.

(S2) *(S1) plus split–merge* Metropolis–Hastings moves. The split proposal is either Jain & Neal's restricted-Gibbs proposal (D2) or an analysable spectral split (P3.A, target C).

(S3) *Lifted, non-reversible* versions of the pairwise moves of Ascolani, Manildo & Zanella (R1), used in P3.E.

From hitting times to mixing times. For lazy reversible chains, the mixing time is equivalent, up to constants, to the largest expected hitting time of a set of stationary mass at least $\alpha < 1/2$ (T20). When the posterior concentrates, $\Pi_n(C_0) \geq 1 - o(1)$, every such set contains C_0. Two consequences follow.

- Upper bounds reduce to bounding $\max_C \mathbb{E}_C \tau_{C_0}$.
- Lower bounds follow from a set $B \not\ni C_0$ that is hard to leave from some start in B. If the posterior flow out of B is small compared with the posterior mass of the start x, then $\mathbb{P}_x(\tau_{B^c} \leq t) \leq t\, Q(B, B^c)/\Pi_n(x)$.

Every project therefore has the same three-step structure: posterior concentration, a bottleneck (or its absence), and a hitting-time bound.

Free-energy profile. For a path that builds a missing block point by point, let $G(m)$ be the log-posterior after m points have been moved, relative to the start. The *barrier* along the path is $-\min_m G(m)$. The free-energy barrier also counts the number of paths, of order $\log \binom{n/K_0}{m} \leq m \log(en/(K_0 m))$. In P3.A the barrier grows with p, and in P3.B with L, while the entropy does not. So for large p or L the energy barrier governs the dynamics.

2.3 P3.A (flagship): an exponential-in-dimension barrier for opening clusters

Working title. *When single-site Gibbs cannot find a missing cluster: dimension-driven metastability of Bayesian clustering with a prior on the number of components.*

Setting. The data are $x_i = \mu_{z_i} + \varepsilon_i \in \mathbb{R}^p$ with $\varepsilon_i \sim N(0, I_p)$ independent.

- *Truth.* A partition C_0 with K_0 blocks, balanced (sizes in $[n/(aK_0), an/K_0]$), with separation $\Delta^2 = \min_{k \neq l} \|\mu_k - \mu_l\|^2/2$ as in P1.
- *Prior.* Block centres $\mu_c \sim N(0, \tau^2 I_p)$, independent and integrated out, with an MFM or DP prior on partitions.
- *Collapsed likelihood.* For a block with n_c points and sum $S_c = \sum_{i \in c} x_i$,

$$\log m(x_c) = -\frac{n_c p}{2} \log(2\pi) - \frac{1}{2} \sum_{i \in c} \|x_i\|^2 + \frac{\tau^2 \|S_c\|^2}{2(1 + n_c \tau^2)} - \frac{p}{2} \log(1 + n_c \tau^2). \tag{1}$$

- *Samplers and regime.* The samplers are (S1) and (S2). The regime is $n, p \to \infty$ with K_0 fixed and $p \geq C \log n$. Both $p \gg n$ and $p \ll n$ are allowed.

Heuristics: two thresholds and a window

Posterior threshold. Take two symmetric clusters with centres $\pm m$, so that $\Delta^2 = 2\|m\|^2$. The log-posterior of C_0 relative to the single block is, from (1),

$$\log \frac{\Pi_n(C_0)}{\Pi_n(\{[n]\})} \approx \frac{n\Delta^2}{4} - \frac{p}{2}\Big(\log \frac{n\tau^2}{4} - 1\Big) - n \log 2, \tag{2}$$

where the last term comes from the DP prior. Hence the posterior prefers two blocks once

$$\Delta^2 > \Delta^2_{\text{post}} \approx \frac{2p}{n}\Big(\log \frac{n\tau^2}{4} - 1\Big) + 4 \log 2.$$

For general K_0 the heuristic scale is $\Delta^2_{\text{post}} \asymp (pK_0/n) \log(n\tau^2/K_0)$, which is a BIC-type Occam penalty. Concentration on the exact partition also needs $\Delta^2 \gtrsim \log n$. The prediction matches simulations closely (Table 6).

Opening threshold. Suppose a block c has merged true clusters, and point x_i belongs to a true cluster whose centre is offset by δ_i from the block mean. Under (S1), the log-odds of moving x_i to a new block rather than leaving it in c are

$$D_i \approx -c(\tau)\, p + \frac{\tau^2 \|\delta_i\|^2}{2(1+\tau^2)} + \log \frac{w_{\text{new}}}{w_c} + O_P\big(\sqrt{p + \|\delta_i\|^2}\big), \qquad c(\tau) = \frac{1}{2}\Big(\log(1+\tau^2) - \frac{\tau^2}{1+\tau^2}\Big) > 0. \tag{3}$$

Here the block mean is taken at the prior mean (Appendix A). The cost $c(\tau)p$ is the price of a p-dimensional centre estimated from *one* point. For two symmetric clusters, $\|\delta\|^2 = \Delta^2/2$, so, ignoring the logarithmic

term, a typical point can open the missing cluster only if

$$\Delta^2 > \Delta^2_{\text{open}} = \frac{4c(\tau)(1+\tau^2)}{\tau^2}\, p \qquad (\approx 0.773\, p \text{ at } \tau^2 = 1).$$

The most extreme of the n points lower this by $O(\sqrt{p \log n})$, a correction that becomes negligible relative to p.

Free-energy profile. Now move m points of the hidden cluster into a new block one at a time. The log-posterior changes by approximately

$$G(m) \approx \frac{\tau^2(m^2\|\delta\|^2 + mp)}{2(1+m\tau^2)} - \frac{p}{2}\log(1+m\tau^2) - m\log n + \log\Gamma(m) + O\Big(\frac{m^2\|\delta\|^2 + pm}{n}\Big), \tag{4}$$

for $m \ll n/K_0$, where the terms $-m\log n + \log\Gamma(m)$ come from the DP prior. This describes a *typical* path, on which the moved points are random members of the hidden cluster. When $\|\delta\|^2 \ll p$, G decreases up to $m^* = p/\|\delta\|^2 - 2/\tau^2$ and increases afterwards. At the minimum,

$$-G(m^*) \approx \frac{p}{2}\log\Big(\frac{\tau^2 p}{\|\delta\|^2} - 1\Big) - p + \frac{2\|\delta\|^2}{\tau^2} + m^*\log n - \log\Gamma(m^*),$$

which is of order p up to a logarithm. At $p = 1000$ and $\tau^2 = 1$ this gives about 650 nats for $\Delta^2 = 100$ and 340 for $\Delta^2 = 200$. A greedy path that moves the most favourable points first (Table 7) has shallower valleys, about half as deep.

The window. Combining the two thresholds, the conjectured picture is

$$\max\Big\{\Delta^2_{\text{post}},\ C\log n\Big\} \ \ll\ \Delta^2\ <\ (1-\epsilon)\,\Delta^2_{\text{open}}.$$

The conjecture is that in this range the posterior concentrates on C_0, single-site moves cannot open the missing cluster in fewer than e^{cp} steps, and one split–merge proposal can. Because the most extreme points lower the opening threshold by $O(\sqrt{p\log n})$, the margin ϵ needs $p \geq C_\epsilon \log n$ with $C_\epsilon \asymp \epsilon^{-2}$. At $p = 1000$ and $n = 400$ single-site moves already open the cluster at between 0.52 and 0.65 of Δ^2_{open}. The window is non-empty as soon as $n \gg K_0 \log n$ and $p \gg \log n$, in both $p \gg n$ and $p \ll n$. In fixed dimension it disappears.

Numerical evidence

Code and settings are in Appendix B. Unless stated otherwise: $n = 400$, $p = 1000$, $K_0 = 2$ symmetric clusters, $\tau^2 = 1$ and a DP(1) prior. Here $\Delta^2_{\text{post}} \approx 21$ and the empirical opening threshold is between 400 and 500.

What the evidence says.

- *The window exists.* For $30 \leq \Delta^2 \leq 400$ ($p = 1000$), the posterior prefers the truth by roughly 900 to 38,000 nats. Yet (S1) never opens the second cluster from the single block, while (S2) finds it in the first sweep.
- *Initialisation matters asymmetrically.* Starts with *large* spurious blocks (4 blocks) reach the truth within a few sweeps, even at $\Delta^2 = 30$. Merging and reorganising existing large blocks is easy, and only *creating* a block is hard. The all-singletons start is not safe either: at $\Delta^2 \leq 50$ it ended in a single block in 11 of 12 runs. Our interpretation, since the runs record only final states, is as follows. The first moves pair points with large noise inner products, of order $\sqrt{p\log n}$. When the signal is weak the resulting blocks are impure and coalesce into one, after which the barrier applies.
- *The thresholds behave as predicted.* The empirical opening threshold, as a fraction of p, rises from ≤ 0.3 to about 0.55 as p goes from 250 to 2000. It approaches the linear prediction $0.773p$ as the $\sqrt{p\log n}$ correction becomes negligible (Table 5). The posterior threshold matches (2) (Table 6).

Targets

Target 2.3.1 (Posterior concentration in high dimension)**.** Let K_0 be fixed, $p \geq C\log n$, and assume a balanced configuration with $\Delta^2 \geq C\log n$.

- If $\Delta^2 \geq (1+\epsilon)\Delta^2_{\text{post}}$, then $\Pi_n(C = C_0) \to 1$ in probability under MFM and DP priors.
- If $\Delta^2 \leq (1-\epsilon)\Delta^2_{\text{post}}$, then $\Pi_n(K_+ < K_0) \to 1$.

Δ^2	$\log \frac{\Pi_n(C_0)}{\Pi_n(\text{one block})}$	(S1) from one block, 200 sweeps	(S2) from one block	(S1) from all singletons	(S1) from 4 large blocks
20	−108	—	stays at one block, which is correct (1/1)	—	—
30	886; 983	stuck (2/2)	truth at sweep 1 (2/2)	collapses to one block (6/6)	truth in 2–5 sweeps (6/6)
50	2874; 2991	stuck (2/2)	sweep 1 (2/2)	one block (5/6); truth (1/6)	2–6 sweeps (6/6)
100	7846; 7999	stuck (2/2)	sweep 1 (2/2)	truth (6/6)	2–5 sweeps (6/6)
200	17791; 17995	stuck (2/2)	sweep 1 (2/2)	truth at sweep 1 (2/2)	1–5 sweeps (6/6)
400	37684; 37962	stuck (2/2)	sweep 1 (2/2)	truth at sweep 1 (2/2)	—
600	57579; 57913	truth at sweep 1 (2/2)	sweep 1 (2/2)	truth at sweep 1 (2/2)	—
1200	117268; 117728	sweep 1 (2/2)	sweep 1 (2/2)	sweep 1 (2/2)	—

Table 4: Hitting the true partition, for $n = 400$ and $p = 1000$. "Stuck" means that no second block of size $\geq n/4$ appeared within the sweep budget. "Truth at sweep s" means that such a block first appeared at sweep s and the run ended with two blocks of size 200. Purity was checked only in the 4-block runs and in the all-singletons runs with $\Delta^2 \leq 150$. (S2) adds two Jain–Neal split–merge proposals per sweep. "4 large blocks" means each true cluster split into random halves, or 4 random blocks mixing both clusters (3 runs each). At $\Delta^2 = 75$ and 150, the all-singletons start found the truth in 6/6 runs. At $\Delta^2 = 800$, (S1) from one block found the truth at sweep 1.

p	250	500	1000	2000
empirical Δ^2_{open}/p	≤ 0.3	0.3–0.4	0.4–0.5	0.5–0.6
typical-point prediction	0.773	0.773	0.773	0.773

Table 5: Opening threshold for (S1) from one block, for $n = 400$: the smallest Δ^2/p on the grid $\{0.3, \ldots, 0.7\}$ at which both runs found the second cluster within 50 sweeps. The ratio rises towards the typical-point value as p grows, as predicted.

p	250	1000	4000
interval where the exact gap changes sign	5–10	20–40	40–80
predicted Δ^2_{post}	7.3	20.8	74.9
exact gaps at the grid points	−263 (5), +228 (10)	−108 (20), +1880 (40)	−3558 (40), +422 (80)

Table 6: Posterior threshold for K, for $n = 400$. Exact log-posterior gaps on the grid $\Delta^2 \in \{p/100, p/50, p/25, p/10\}$, with the value of Δ^2 in brackets.

Here $\Delta^2_{\text{post}} \asymp (pK_0/n)\log(n\tau^2/K_0)$, with the sharp constant for symmetric configurations.

This is the statistical half, and it is of independent interest. It gives a high-dimensional Bayesian threshold for the number of clusters. When $p \gg n$ it sits far above the spectral and information-theoretic thresholds of P1, which are of order $\sqrt{pK_0^2/n}$. So the conjugate posterior is a conservative counter of clusters in high dimension, which links P3 back to P1. Chandra et al. (C7) show degeneracy when n is fixed. The finite-dimensional analogue is C2. For the DP prior the target goes against Miller & Harrison's inconsistency (C1), and this is deliberate. That inconsistency is driven by tiny extra clusters, which cost only $O(\log n)$ in fixed dimension. Here a singleton costs $c(\tau)p \gg \log n$, so DP and MFM should behave alike when $p \gg \log n$. In fixed dimension the DP half of the target would fail.

Target 2.3.2 (Exponential lower bound for single-site moves)**.** Let $\Delta^2 \leq (1-\epsilon)\Delta^2_{\text{open}}$ and $p \geq C_\epsilon \log n$. With high probability over the data, start (S1) from any *coarsening* of C_0 with fewer than K_0 blocks, that is, a partition whose blocks are unions of true clusters, in particular the single block. It then needs at least $\exp(c_\epsilon p)$ updates before it visits a partition with K_0 blocks of size at least ηn. Together with Target 2.3.1, in the window $t_{\text{mix}}(\text{S1}) \geq e^{cp}$ while $\Pi_n(C_0) \to 1$. Starts outside this metastable well behave differently. Moves among *existing* large blocks are not obstructed, and the all-singletons start often escapes (Table 4). There the posterior mass of the start is tiny, so the bottleneck bound of Section 2.2 is vacuous.

	$m=1$	2	4	8	16	32	64	128	200	barrier
$p=1000$, $\Delta^2=100$	−61	−126	−222	−307	−303	−15	868	3458	7756	325
$p=1000$, $\Delta^2=200$	−45	−75	−112	−68	182	978	2973	8433	17673	112
$p=1000$, $\Delta^2=600$	14	82	254	744	1952	4720	11085	27891	57391	none
$p=1000$, $\Delta^2=1200$	100	306	784	1927	4507	10145	22948	56684	117010	none
$p=2$, $\Delta^2=8$	1	4	13	25	50	100	194	366	533	none

Table 7: The exact profile $G(m)$, in nats, along a greedy single-site path from one block to the truth. At each step the point of the hidden cluster that most increases the posterior is moved. The barrier is the depth of the valley over all m: 325 at $m=12$ for $\Delta^2=100$, and 112 at $m=5$ for $\Delta^2=200$. Below the empirical opening threshold (400–500 here), the path must cross a valley of depth 110–325 nats, which was never crossed in the runs. Above it, or in dimension 2, the path is uphill from the first step.

Target 2.3.3 (Polynomial mixing with split–merge)**.** In the window, (S2) with a suitable split proposal has $t_{\text{mix}} \le \text{poly}(n,p)$ with high probability. Two split proposals are of interest. One is a randomised spectral split of a block, based on the sign of the top eigenvector of its centred Gram matrix, with independent random flips so that the proposal probability is explicit. The other, the practical target, is Jain–Neal's restricted-Gibbs proposal.

Target 2.3.4 (Initialisations and fixed dimension)**.** • Take $K_0=2$ and a partition with two blocks of size $\ge cn$ whose majorities come from different true clusters. From any such partition, (S1) hits C_0 in polynomially many sweeps. In the simulations, even 4 random blocks were enough. For $K_0 \ge 3$ the right condition is unclear, because the property can be lost during the run: two spurious halves of one cluster may merge before another merged pair is separated.

- For fixed p and $\Delta^2 \ge C\log n$, (S1) hits C_0 from any start in $O(n^2 \,\text{polylog}\, n)$ updates. The slowest stage is merging an even spurious split, which is a nearly neutral Moran dynamics (P3.E).

Proof route

1. *Target 2.3.1.* Use the exact formula (1). Write the log-posterior ratio between C and C_0 as a quadratic form in the noise. Control it uniformly over partitions with Hanson–Wright and Gaussian-chaos bounds plus a union bound, organised by the distance to C_0 (the approach of EGV and of the variable-selection literature, V1).
2. *Target 2.3.2.* Take as bottleneck B the set of partitions with no block containing at least m^* points of a hidden cluster and few others, with m^* the minimiser of (4). Bound $\Pi_n(\partial B)/\Pi_n(\text{start}) \le e^{-cp}$. This needs (4) uniformly over subsets of size $\le m^*$ (chaos bounds plus a union bound costing $m^*\log n \ll cp$) and a Peierls-type summation over the rest of the partition. Then use $\mathbb{P}_x(\tau_{B^c} \le t) \le tQ(B,B^c)/\Pi_n(x)$ (Section 2.2). Tosh & Dasgupta (L1) and Ben Arous–Wein–Zadik (H3) are the templates.
3. *Target 2.3.3.* Show "unimodality through local moves". Every partition other than C_0 should have a split, merge or single-site neighbour with posterior ratio $\ge n^c$, proposed with probability $\ge n^{-c'}$. Then apply Zhou & Chang's bound (V3) with $\log(1/\pi_{\min}) = O(np)$, or a drift argument for $\max_C \mathbb{E}_C\tau_{C_0}$ together with T20. For the spectral split, the window lies above the spectral detection threshold, because $(pK_0/n)\log n \gg \sqrt{pK_0^2/n}$ when $p \gg n$ (and $\Delta^2 \gg \log n$ when $p \lesssim n$).

What is new.

- The first quantitative result, upper or lower, for a random-K sampler that people actually use.
- A theorem behind the folklore that split–merge moves are needed (S7, S9, D2, D6), with an explicit mechanism and explicit thresholds.
- Tosh & Dasgupta's bottleneck comes from separation with K fixed. Here it comes from dimension and from the move that changes K.
- The same mechanism covers overfitted mixtures with K fixed and empty components (C3, C8), because re-populating an empty component is the same move.

Practical message. In high dimension, a single-site sampler with a prior on K cannot be trusted to discover clusters. Neither the single-block start nor the all-singletons start is safe. Starts that give each cluster a

large block of its own are safe, for instance over-clustering by K-means. Split–merge moves are needed. The posterior itself is fine.

Risks.

- The bottleneck in Target 2.3.2 must block every single-site route, including routes through many small blocks. The Peierls summation is the delicate step.
- Analysing the restricted-Gibbs split proposal may be hard. The fallback is the spectral proposal.
- MFM weights involve ratios of $V_n(t)$, which only add $O(\log n)$ terms, so they should be harmless.
- Competition: none found, but the area is active (R1).

Venue. AoS or JRSS-B for Targets 2.3.1–2.3.3. Target 2.3.1 alone would suit Biometrika or Bernoulli.

2.4 P3.B: layers as dimensions — the same barrier in sparse multilayer SBMs

Working title. *Single-site samplers cannot find communities in sparse multilayer networks with an unknown number of communities.*

Setting. Take the sparse multilayer SBM of P4: n nodes, L layers, K_0 communities and $B^{(\ell)} = \rho_n B_0^{(\ell)}$ with $n\rho_n = O(1)$. Use an MFM or DP prior on partitions and independent Beta(a, b) priors on every $B_{kl}^{(\ell)}$. Integrating the Beta variables out gives a Beta–Bernoulli likelihood. The samplers are (S1), and (S2) with a spectral split proposal built from the bias-adjusted sum of squares $S = \sum_\ell \{(A^{(\ell)})^2 - D^{(\ell)}\}$ of P4.

Heuristic. Opening a singleton block $\{i\}$ out of a block M has three effects.

- *Cost.* It creates K new block-pair parameters in each layer, KL in all. Each is estimated from about n/K trials with $O(n\rho_n/K)$ expected successes. Under a Beta prior each costs about $\log(n/K)$ when sparse, and $\frac{1}{2}\log(n/K)$ when dense.
- *Gain.* A single node gains at most its own information, of order $L\, n\rho_n I_1$, where I_1 is the per-layer, per-node divergence. In homogeneous layers the gain is zero: a singleton's edges to M are exactly as frequent as M's internal edges.
- *Net effect.* The log-odds of opening are therefore about $-L\,[\,c\,K\log(n/K) - O(n\rho_n I_1)\,] - \log(n/\alpha)$. They are *linear in* L and negative once $n\rho_n \ll K\log n$, which is the sparse regime of P4.

The gain from a complete block grows like $L\,n \cdot n\rho_n$, so the posterior favours the truth once L is large enough, roughly $Ln\rho_n I \gtrsim \log n$. The number of layers L plays the role of p in P3.A.

L	singleton log-odds, mean (max)	greedy-path barrier	$\log \frac{\Pi_n(C_0)}{\Pi_n(\text{one block})}$	(S1) from one block, 40 sweeps
1	-10.2 (-7.3)	—	-180	—
5	-26.9 (-21.1)	85	$+139$; $+111$	stuck (2/2)
20	-89.0 (-73.1)	197	$+1367$; $+1363$	stuck (2/2)
60	-256.0 (-242.6)	524	$+4752$; $+4768$	not run

Table 8: Multilayer SBM with $n = 400$, $K_0 = 2$ and $n\rho_n = 2$ per layer. Each layer is assortative or disassortative at random, with rates $2\rho_n$ and $0.5\rho_n$. The prior is DP(1) with Beta(1,1). The two gap values come from two independent draws, the barrier (the depth of the valley over all m) from the second. Started from the truth, (S1) stays near the truth, with final block sizes 202/198 and 206/194 for $L = 5$.

Evidence. Table 8. The cost of a singleton grows by about 4.2 nats per layer, and the greedy-path barrier grows roughly linearly in L. The posterior prefers the truth by a margin that also grows linearly in L. Single-site Gibbs started from one block never found the second community.

Targets.

(B1) *Posterior concentration.* With random K in the sparse multilayer SBM, the posterior concentrates on (K_0, C_0) under $Ln\rho_n I \geq (1+\epsilon)\log n$, together with a condition that excludes cancellation. This is new even with K known for the Bayesian posterior.

(B2) *Lower bound.* From under-clustered starts, (S1) needs at least $\exp(cL\log n)$ updates when $n\rho_n \ll K\log n$.

(B3) *Upper bound.* (S2) mixes in polynomial time with split proposals given by the leading eigenvectors of the restriction of S to a block. This needs P4's control of the noise edge of S, so it would be a sampler-level application of P4.

What is new. Barriers for random-K samplers on networks, with a mechanism, the number of new block parameters, that grows with the number of layers. It also suggests why single-network samplers that keep many groups underestimate k (S9): the cost of opening a block grows with the current number of blocks K (Section 2.5).

Risks.

- Posterior concentration for sparse Bayesian SBMs with Beta priors is delicate.
- High-degree vertices make the noise edge of S hard to control (P4, Q1).
- The homogeneous case (no single-node signal) and the heterogeneous case need separate arguments.

Venue. AoS or JASA, as a companion to P4. It could also be a section of the P3.A paper ("a second example of the mechanism").

2.5 P3.C: the number of communities acts as a dimension

Working title. *Why single-site samplers underestimate the number of communities: polynomial barriers of growing degree in Bayesian stochastic block models.*

Why this is more than an extension of Zhuo & Gao. Rapid mixing near the truth with random K (now project C3 below) is natural but incremental. The simulations below show something stronger. In a collapsed Bayesian SBM, single-site samplers cannot *create* communities, and the obstruction grows with the number of communities. This makes a documented failure precise. Riolo et al. (S9) report that their sampler "appears not to equilibrate fully when the number of groups becomes large" and underestimates k. Peixoto (S7) finds single-node moves "effectively non-ergodic". No theorem explains either observation.

Setting. A planted-partition SBM with K_0 communities of size n/K_0, within-community probability p_n and between-community probability q_n. Two regimes are of interest: dense (p, q constant) and exact recovery ($p, q \asymp \log n/n$). The prior is DP or MFM on partitions, with independent Beta(a, b) priors on all block-pair probabilities, integrated out (as in S2, S10, S13). The samplers are (S1), single-node collapsed Gibbs with a new-block option, and (S2), (S1) plus split–merge.

Mechanism. Suppose two communities H and R sit in one block M, and the other $K-2$ communities form blocks of their own. Moving a node $i \in H$ into a new singleton block has three effects.

- *Cost.* It creates about K new block-pair parameters, between $\{i\}$ and each existing block. Each is estimated from about n/K trials, so under a Beta prior each costs about $\frac{1}{2}\log(n/K)$ in the dense regime and about $\log(n/K)$ in the sparse regime (Appendix A, A.6).
- *Gain.* None. In the planted partition model, i's connection rate to M equals M's internal average rate, and its rates to the other blocks equal M's.
- *Prior.* The partition prior charges about $\log(n_M/\alpha)$.

So the log-odds of opening are about $-\frac{K}{2}\log(n/K) - \log(n/\alpha)$, *linear in* K. A gain appears only once a nucleus of $m \gtrsim \sqrt{K\log(n/K)/\mathrm{KL}(p \,\|\, \bar{p})}$ nodes of H sits together. Here $\bar{p}$ is the average rate inside M. Meanwhile the DP prior's preferential attachment charges about $\log(n_M/m)$ for each added node, partly offset by the number of candidates. The heuristic hitting time is therefore polynomial, $n^{(K/2)(1+o(1))}$, with a degree that grows with K. It is superpolynomial if $K \to \infty$.

The common principle with P3.A and P3.B. In each case the cost of opening a block is the number of new parameters times the cost of estimating each from what a single unit provides.

- *Gaussian mixtures:* a p-dimensional centre estimated from one point, cost $c(\tau)p$.
- *SBM:* a K-dimensional row of the connectivity matrix estimated from n/K edges per entry, cost about $\frac{K}{2}\log(n/K)$.
- *Multilayer SBM:* KL such entries.

Evidence. Table 9.

- *The cost grows linearly in* K_0, by about 1.9 nats per extra community, and so does the barrier.

K_0	n/K_0	log-odds of opening, mean (max)	greedy barrier (m)	$\log \frac{\Pi_n(C_0)}{\Pi_n(\text{merged})}$	(S1) from the merged start, 30 sweeps	(S1) from one block, 100 sweeps: large blocks found
2	400	-9.5 (-5.5)	12.5 (6)	44753	found in 1/2 runs (sweep 4)	same as merged start
4	200	-14.2 (-9.4)	17.3 (4)	11153	stuck (2/2)	2 and 3 of 4
8	100	-22.0 (-15.5)	23.1 (7)	2682	stuck (2/2)	1 and 2 of 8
16	50	-36.1 (-29.9)	45.4 (8)	522	stuck (2/2)	1 and 2 of 16

Table 9: Planted-partition SBM with $n = 800$, $p = 0.5$, $q = 0.05$, a DP(1) prior and Beta(1,1) priors. The "merged start" is the truth with two communities merged. The log-odds are for moving one node of the hidden community into a new block. The barrier is the depth of the greedy single-site path that rebuilds the hidden community. Started from the truth, (S1) stayed there in all runs.

- *The posterior prefers the truth by 500 to 45,000 nats.* Yet from the merged start (S1) separated the two communities only once, for $K_0 = 2$.
- *Underestimation.* From the default one-block start, 80,000 single-site updates found only 1 to 3 of the communities, whatever K_0.
- *No spurious blocks at the truth.* The total odds of an extra singleton block are $e^{-3.2}$, $e^{-7.0}$, $e^{-13.9}$ and $e^{-26.0}$ for $K_0 = 2, 4, 8, 16$.

Three projects, and an umbrella.

(C1) **Core paper: single-site samplers cannot discover communities.** *Novelty high, risk medium, venue AoS, JASA or JRSS-B.*

(a) *Lower bound.* From any coarsening of the truth that merges two communities, (S1) needs at least $(n/K_0)^{cK_0}$ updates to separate them. The bound is superpolynomial when $K_0 \to \infty$; for instance $K_0 \asymp \log n$ gives $n^{c \log n}$.

(b) *Underestimation law.* From one block, after T updates (S1) has at most $O(\log T / \log(n/K_0)) + O(1)$ blocks of size $\geq \epsilon n/K_0$, with high probability. With a polynomial budget the number of communities found stays bounded while K_0 grows, as in Table 9 and in S9.

(c) *Upper bounds.* (S2) with a spectral split proposal hits the truth in $\text{poly}(n, K_0)$ steps, and near the truth (S1) mixes fast (C3).

(d) *Sharpness.* Identify the exponent for fixed K_0: heuristically about $K_0/2$, plus the cost of building the nucleus.

The proof route is the architecture of P3.A. Posterior concentration (C2) comes first. The bottleneck is the set of partitions with no block containing m^* or more nodes of the hidden community. The collapsed likelihood is a product of Beta functions of edge counts, which concentrate by Bernstein bounds, so this is technically *cleaner* than the Gaussian chaos of P3.A. The rest is a union bound over nuclei and the escape bound of Section 2.2.

(C2) **Statistical companion: how many communities does the posterior see?** *Novelty medium to high, risk low to medium, venue Biometrika, Bernoulli or AoS.*

(a) Posterior concentration on (K_0, C_0) in the exact-recovery regime with standard Beta priors and $K_0 = K_n \to \infty$. Geng et al. (S10) cover only $K_0 \in \{2, 3\}$; Jiang & Tokdar (S14) need an assortative prior.

(b) *DP priors are consistent for the number of communities.* This is a conjecture that contrasts with mixtures (C1). An extra block costs a factor of about $n^{-K/2} \cdot n^{-1}$, while there are only about n ways to start one. In fixed-dimensional mixtures the analogous cost is $O(1) \cdot n^{-1}$, which is why DP mixtures keep tiny spurious clusters there. The odds above support this. No such result was found in the literature searched.

(C3) **Stepping stone: local mixing with random** K**.** *Novelty medium, risk low, venue JMLR or Bernoulli.* This is Zhuo & Gao (U1) with create and empty moves and MFM or DP weights, without restriction to balanced partitions.

- From a warm start with K_0 blocks and misclassification $o(1)$, $\tau_\epsilon \lesssim n^2(n^2 I + \log \epsilon^{-1})$.
- Spurious small blocks are emptied in $O(n \log n)$ steps, because opening one costs about $n^{-K/2}$.

Together with (C1) this gives the full picture: fast locally, slow at creating communities, and polynomial globally once split–merge moves are added. It makes a good first chapter.

(C4) **Umbrella (optional, ambitious): an opening-cost principle.** *Novelty high, risk high, venue AoS.* A general lower bound for single-site samplers on collapsed conjugate exponential-family partition posteriors. The log-odds of opening a block from one unit are at most minus the information cost of the new parameters, plus the unit's own information. The corollaries would be e^{cp} for Gaussian mixtures (P3.A), n^{cK} for SBMs (C1), n^{cKL} for multilayer SBMs (P3.B), and $n^{c(K_r+K_c)}$ for latent block models (S17).

Risks.

- For small K_0 the barrier is moderate: $K_0 = 2$ was crossed in one of two runs. So (C1)(a) is most striking for growing K_0.
- In degree-corrected SBMs, which Riolo et al. and Peixoto use, degree parameters change the costs.
- The nucleus phase combines likelihood, the DP prior's preferential attachment and entropy. It must be controlled uniformly.
- With MFM weights, the ratio $V_n(t+1)/V_n(t)$ enters the cost.

2.6 P3.D: which random-K samplers are geometrically ergodic?

Working title. *The tail of the prior on K decides geometric ergodicity of trans-dimensional mixture samplers.*

Setting. An MFM with $K \sim p_K$ on $\{1, 2, \dots\}$, unbounded. The weights are symmetric $\mathrm{Dir}_K(\gamma)$ ("static") or $\mathrm{Dir}_K(\alpha/K)$ ("dynamic", M9). The components are Gaussian: conjugate, or as in Qin's example (T13). Four classes of sampler are compared:

(a) reversible jump with birth/death or split/merge moves (M1, T13), where K changes by ± 1;

(b) allocation samplers with add/remove-empty moves (M5, S3, S12), also ± 1;

(c) the telescoping sampler (M9), where K is drawn exactly given the partition;

(d) collapsed partition samplers (M8), where the state space is finite.

Proposition 2.6.1 (the posterior of K has the prior's tail; our computation)**.** Fix the data. Then $m(x \mid K = k) \to m_\infty(x) \in (0, \infty)$ as $k \to \infty$, so $\Pi(K = k \mid x) \asymp p_K(k)$.

Sketch. Under the static MFM, $P(C \mid K = k) = k_{(t)} \prod_c \gamma^{(n_c)} / (\gamma k)^{(n)}$, which tends to 1 for the all-singletons partition and to 0 otherwise. So $m_\infty = \prod_i m(x_i)$. Under the dynamic MFM, $P(C \mid K = k)$ tends to the CRP(α) probability of C, so m_∞ is the DP-mixture marginal. □

Proposition 2.6.2 (bounded jumps in K and heavy tails; our observation)**.** Suppose $\Pi(K \geq k \mid x) \geq c\, k^{-a}$ for some $a > 0$, and the sampler changes K by at most J per step. Then the sampler is not geometrically ergodic.

Sketch. Suppose the sampler were geometrically ergodic.

- There would be a drift function $V \geq 1$ with $\pi(V) < \infty$ and $PV \leq \lambda V$ outside a small set C.
- Since $V \geq 1$, a supermartingale argument gives $V(x) \geq \mathbb{E}_x \lambda^{-\tau_C}$.
- Small sets have bounded K-range, because $P^m(x, \cdot)$ only reaches K within mJ of $K(x)$.
- So $\tau_C \geq (K(x) - k_C)/J$, which would force $\pi(\lambda^{-K/J}) < \infty$. This contradicts the polynomial tail. It is the argument of Jarner & Tweedie (T19). □

Consequence. Under the beta-negative-binomial $\mathrm{BNB}(1, 4, 3)$ prior used in M9, whose probabilities decay like k^{-5} so that $\Pi(K \geq k) \asymp k^{-4}$, or Gnedin-type priors (S15, S13), every sampler of classes (a) and (b) fails to be geometrically ergodic. The telescoping sampler avoids this by design, and Poisson or geometric priors avoid it too.

Targets.

(D3) *Geometric ergodicity with unbounded K under light tails*, $p_K(k+1)/p_K(k) \to r \in [0,1)$, for classes (a) and (b). This extends Qin's Proposition 13 beyond $k_{\max}$. Take a drift function $V = e^{sK} \times$ (within-model drift). When $K > n$ there are always at least $K - n$ empty components, so death moves are available and the drift in K is explicit. Then use Qin's countable-model Theorem 7 or a direct drift-and-minorisation argument.

(D4) *Polynomial ergodicity under polynomial tails*, with the rate determined by the tail exponent a, via polynomial drift conditions.

(D5) *Uniform ergodicity of the telescoping sampler for every* p_K, with conjugate components and fixed hyperparameters. If the update of (number of components, weights, parameters) given the partition is an exact draw from its conditional, the partition sequence is a finite Markov chain that de-initialises the rest (T6, T7). The same check applies to Argiento & De Iorio's conditional sampler (M10).

(D6) *Harris recurrence* of reversible-jump mixture samplers from every start, via T3 (Theorem 20 or 21). This closes the "π-almost every start" gap of Route 3.

What is new. A complete qualitative classification of random-K samplers.

- Collapsed and exact-K samplers are uniformly ergodic.
- Samplers that move K by ± 1 cannot be geometrically ergodic under polynomial tails (Proposition 2.6.2), and should be under light tails (target (D3)). Together these would give an almost complete dichotomy.

The practical message is that a heavy-tailed prior on K should be paired with exact updates of K (telescoping) or with collapsed samplers.

Risks. Low. (D3) requires within-model drift uniformly in K when there are continuous parameters. Qin's Lemma 12, that the within-model kernel depends only on the allocations, is the key device.

Venue. Biometrika (a short paper with the clean dichotomy), Bernoulli, Statistics and Computing or JCGS. A good first project: short, low-risk and useful.

2.7 P3.E: posterior scaling limits in low dimension

Working title. *Moran dynamics with selection: the posterior scaling limit of collapsed Gibbs for random partitions.*

Setting. Fixed p, K_0 clusters with $\Delta^2 \geq C \log n$, and a DP, MFM or Pitman–Yor prior. The samplers are (S1), the lifted non-reversible sampler (S3) of R1, and (S2).

Observations.

- *With no data*, random-scan (S1) for CRP(α) is the infinite-alleles Moran model (P2). For fixed α its time scale is $\Theta(n^2)$ updates.
- *With well-separated clusters in fixed dimension*, the likelihood is nearly flat across sub-partitions of one true cluster, except for the Occam factor. Splitting a true cluster of size m into sizes a and $m - a$ costs about $\frac{p}{2} \log(a(m-a)\tau^2/m)$.
- *Together*, the sub-cluster frequency $x = a/m$ then behaves like a Moran model with *selection towards fixation*. On the time scale of m^2 updates its drift is about $-\frac{p}{2}(1-2x)$, pushing x away from $\frac{1}{2}$, while the Moran part contributes $\sqrt{x(1-x)}\, dW$.

Targets.

(E1) A diffusion limit, of Wright–Fisher type with selection, for (S1) started from a spurious split of a true cluster.

(E2) A worst-case mixing time of $\Theta(n^2)$ updates for (S1) and $\Theta(n)$ for the lifted sampler (S3). This would be the *posterior* counterpart of Ascolani–Manildo–Zanella's prior-case comparison, which their 2026 slides list as open.

(E3) How DP, MFM and PY(σ) priors change the mutation and selection terms. Under PY with $\sigma > 0$, new blocks are created at a rate that grows with the number of blocks, which may change the picture qualitatively.

Risks. The authors of R1 are working in this direction, so the competition is real. Diffusion limits with selection that is singular at the boundary are technical. In high dimension the P3.A barrier dominates, so this is a low-dimensional story.

Venue. AoS, AAP or Bernoulli.

2.8 P3.F: does tempering remove the barrier?

Working title. *Tempering does not help Bayesian clustering find missing clusters in high dimension.*

Setting. As in P3.A, with tempered posteriors $\pi_\beta(C) \propto p(C)\prod_c m_\beta(x_c)$: the likelihood is raised to the power β and the partition prior is not tempered. Parallel or simulated tempering runs on a ladder $\beta_1 < \cdots < \beta_J = 1$, with (S1) inside each temperature.

Heuristic. Tempering by β is equivalent to scaling the data by $\sqrt{\beta}$ and τ^2 by β. In the original units the opening threshold becomes

$$\Delta^2_{\text{open}}(\beta) = \frac{4\,c(\sqrt{\beta}\,\tau)\,(1+\beta\tau^2)}{\beta^2\tau^2}\,p,$$

which *increases* as β decreases. At $\tau^2 = 1$ it is $0.773p$ at $\beta = 1$, $0.866p$ at 0.5, $0.968p$ at 0.1, and tends to $\tau^2 p$ as $\beta \to 0$. Meanwhile the separated state is favoured only above a critical β_c. So in this heuristic, raising the temperature (lowering β) does not help open the missing cluster, and at β_c the barrier is still of order p. The profiles in Table 10 agree, for one data set. This is the first-order-transition scenario that makes tempering torpid, as for the mean-field Potts model (E2).

β	1	0.5	0.2	0.1	0.05	0.03
greedy-path barrier (nats)	112	384	569	636	636	599
location m of the minimum	5	12	33	58	91	114
$\log \pi_\beta(C_0)/\pi_\beta(\text{one block})$	17673	7836	2177	472	−230	−419

Table 10: Tempered profiles for $n = 400$, $p = 1000$, $\Delta^2 = 200$ and $\tau^2 = 1$. The barrier grows as β decreases from 1 to 0.1 and stays near 600–640 nats below that. It is 636 nats where the preferred state switches, between $\beta = 0.1$ and 0.05.

Targets.

(F1) In the window of P3.A, parallel and simulated tempering based on (S1), with any ladder of polynomially many temperatures, have spectral gap at most e^{-cp}. The tool is the persistence criterion of E2.

(F2) Tempering combined with split–merge is polynomial.

(F3) Do other annealing schemes avoid the transition? Candidates are tempering the prior (larger α), data or subsample annealing (L2), and annealing τ^2.

What is new. This rules out the most common generic remedy, with a clear mechanism, and connects clustering posteriors to the theory of first-order transitions and torpid tempering.

Venue. AAP or AoS, possibly merged into P3.A as a section.

2.9 Further directions

- **Non-conjugate samplers.** In Neal's Algorithm 8, the auxiliary components' parameters are drawn from the prior. In high dimension a prior draw lies at distance about $\tau\sqrt{p}$ from every data point. The chance that one of m auxiliary draws beats the current block is then about $m\,e^{-c\tau^2 p}$, whatever the signal, so the barrier of P3.A is even stronger. Target: the same theorems with an explicit dependence on m.
- **Hierarchical priors.** With unknown covariances (normal–inverse-Wishart), the Occam penalty of a complete block is of order $p^2 \log n$, which moves the posterior threshold up. The cost of opening a singleton stays of order p, but its constant depends on the prior scale matrix. How the window changes has to be worked out. It relates to the high-dimensional degeneracy in C7.
- **Slow mixing of the actual Bayesian SBM posterior with K known.** Below the Kesten–Stigum threshold, or with $k \geq 5$, prove an overlap gap or a free-energy well for the posterior itself rather than for modularity (L4) or surrogate Ising measures (U6). This is a probability or TCS project (AAP, PTRF, FOCS).
- **Random-K degree-corrected and nested SBMs.** Quantify Peixoto's observation (S7) that single-node moves are "effectively non-ergodic" on small networks, by the opening cost of P3.C with degree correction.

- **Coupling-based diagnostics with guarantees.** Combine D10 with the bottleneck sets of P3.A. This would detect non-convergence provably, through coupled chains that fail to meet.

2.10 Recommended route

1. **First paper (3–4 months): P3.D.** Propositions 2.6.1–2.6.2 are essentially proved above, (D5) is a duality argument, and (D3) is a drift computation. The result is a clean dichotomy with a practical message, and it builds the drift and duality machinery for later work.
2. **Main paper (12–18 months): P3.A.** Prove Target 2.3.1 first, since it is statistical and publishable on its own. Then do Target 2.3.2 (bottleneck) and Target 2.3.3 (spectral split with Zhou–Chang). Include the simulations of Tables 4–7, and add Table 10 (P3.F) as a section showing that tempering does not help.
3. **Follow-up: P3.B with P4.** The multilayer version reuses the P3.A architecture. The split proposal comes from P4's statistic S, so progress on P4's noise edge feeds directly into (B3).
4. **Alternatives.** For a student who prefers networks, P3.C replaces P3.A as the main paper. Its stepping stone (C3) is a low-risk first paper, and its core result (C1) is a strong second. P3.E is attractive but competitive. Contact with the authors of R1 may be wiser than racing them.

Checks before starting.

- Read the conductance sets in Tosh & Dasgupta (L1), the canonical paths in Zhuo & Gao (U1), Qin's Section 4.3 and Theorem 7 (T13), and Zhou & Chang's general bound (V3).
- Obtain the supplement of R1, version 2, to see exactly how far Theorem 4 goes for random partitions.
- Run a final search for 2026 preprints. Suggested terms: "Dirichlet process mixture Gibbs high-dimensional new cluster", "split–merge mixing time", "telescoping sampler ergodicity", and "multilayer SBM MCMC unknown number of communities".
- Check whether the identification of prior-only collapsed Gibbs with the infinite-alleles Moran model (P2) is already stated in the literature.
- Compare the conditions of Chandra, Canale & Dunson (C7) with the regime of Target 2.3.1.

A Heuristic derivations

A.1 Collapsed quantities. For the model of P3.A, take a block c with n_c points and sum S_c, and set $v_c = \tau^2/(1+n_c\tau^2)$. The predictive law of a new point is $N(v_c S_c, (1+v_c)I_p)$, and for an empty block it is $N(0,(1+\tau^2)I_p)$. Formula (1) follows by completing the square.

A.2 Singleton log-odds (3). Let $n_c \gg 1$, so that $v_c S_c \approx \bar{x}_c$ and $v_c \approx 0$. Then

$$D = \log\frac{w_{\text{new}}}{w_c} - \frac{p}{2}\log(1+\tau^2) - \frac{\|x\|^2}{2(1+\tau^2)} + \frac{\|x-\bar{x}_c\|^2}{2}.$$

Write $x = \bar{x}_c + \delta + \varepsilon$. When $\bar{x}_c = 0$, both $\|x\|^2$ and $\|x-\bar{x}_c\|^2$ equal $\|\delta+\varepsilon\|^2 = p + \|\delta\|^2 + O_P(\sqrt{p+\|\delta\|^2})$. Since $\frac{1}{2} - \frac{1}{2(1+\tau^2)} = \frac{\tau^2}{2(1+\tau^2)}$, this gives (3) with $c(\tau) = \frac{1}{2}[\log(1+\tau^2) - \tau^2/(1+\tau^2)]$. If $\bar{x}_c \neq 0$, an extra term $-(\|\bar{x}_c+\delta\|^2 - \|\delta\|^2)/(2(1+\tau^2))$ appears, which makes opening harder for blocks far from the prior mean. In the symmetric case $\|\delta\|^2 = \Delta^2/2$, which gives Δ^2_{open}. The fluctuation term has standard deviation about $\frac{\tau^2}{2(1+\tau^2)}\sqrt{2p+4\|\delta\|^2}$, so the best of n points gains about $\sqrt{2\log n}$ times that. This explains the downward shift of the empirical threshold in Table 5.

A.3 Profile (4). Move a set B of $m \ll n_c$ hidden points out of a block with mean about 0. The block term of (1) for B contributes $\tau^2\|S_B\|^2/(2(1+m\tau^2)) - \frac{p}{2}\log(1+m\tau^2)$, with $\|S_B\|^2 \approx m^2\|\delta\|^2 + mp$. The change in the remaining block is $O(m^2\|\delta\|^2/n_c + pm/n_c)$, and the DP prior contributes $\log\alpha + \log\Gamma(m) + \log\Gamma(n_c - m) - \log\Gamma(n_c)$.

- *Minimiser.* The derivative of the likelihood part is $\frac{\tau^2 m}{2(1+m\tau^2)^2}\left[2\|\delta\|^2 + m\|\delta\|^2\tau^2 - p\tau^2\right]$, which vanishes at $m^* = p/\|\delta\|^2 - 2/\tau^2$.
- *Barrier.* Substituting m^* into the approximate likelihood part gives exactly

$$p - \frac{2\|\delta\|^2}{\tau^2} - \frac{p}{2}\log\Big(\frac{\tau^2 p}{\|\delta\|^2} - 1\Big).$$

  Adding the prior part $-m^*\log n + \log\Gamma(m^*)$ gives the barrier formula of Section 2.3.

A.4 Posterior gap (2). Two blocks of size $n/2$ around $\pm m$ each contribute about $\frac{n}{4}\|m\|^2 + \frac{p}{2} - \frac{p}{2}\log(n\tau^2/2)$ through the last two terms of (1). The merged block contributes about $\frac{p}{2} - \frac{p}{2}\log(n\tau^2)$, because the centres cancel. The difference is $\frac{n}{2}\|m\|^2 + \frac{p}{2} - \frac{p}{2}\log(n\tau^2/4)$. The DP prior adds $\log\{\alpha\Gamma(n/2)^2/\Gamma(n)\} \approx -n\log 2$. Substituting $\|m\|^2 = \Delta^2/2$ gives (2).

A.5 Tempering. Since $N(x;\mu,I)^\beta \propto N(x;\mu,I/\beta)$, with a constant that does not depend on the partition, tempering amounts to replacing x by $\sqrt{\beta}\,x$ and τ^2 by $\beta\tau^2$. The effective separation is $\beta\Delta^2$. Inserting this into Δ^2_{open} and converting back gives $\Delta^2_{\text{open}}(\beta)$. As $\beta \to 0$, $c(\sqrt{\beta}\tau) \approx \beta^2\tau^4/4$, so $\Delta^2_{\text{open}}(\beta) \to \tau^2 p$.

A.6 Cost of a new block-pair parameter. *Dense regime.* With N trials and a rate bounded away from 0 and 1, the Laplace approximation of the Beta–Bernoulli marginal gives the plug-in likelihood times a factor of order $N^{-1/2}$. The cost is therefore $\frac{1}{2}\log N + O(1)$ per parameter, which is the $\frac{1}{2}\log(n/K)$ used in P3.C. *Sparse regime.* With a Beta(1,1) prior and N trials with $s = O(1)$ successes, the Beta–Bernoulli marginal is $s!(N-s)!/(N+1)!$. Compared with a known rate λ/N, its log is $-\log N + \log s! - s\log\lambda + \lambda = -\log N + O(1)$. So each new block-pair parameter costs about $\log(n/K)$ in the sparse regime. A singleton block adds K such parameters per layer.

A.7 Proposition 2.6.1. With $P(C \mid k) = k_{(t)}\prod_c \gamma^{(n_c)}/(\gamma k)^{(n)}$,

$$k_{(t)}/(\gamma k)^{(n)} \sim \gamma^{-n}k^{t-n} \to 0 \quad \text{unless } t = n,$$

and $\prod_c \gamma^{(1)} = \gamma^n$ for the all-singletons partition. Under Dir(α/k) weights, $(\alpha/k)^{(n_c)} \sim (\alpha/k)(n_c-1)!$. Hence $P(C \mid k) \to \alpha^t\prod_c(n_c-1)!/\alpha^{(n)}$.

B Simulation details

All simulations are in Python with NumPy and SciPy. They are short and were run on two cores, and each table took under a minute except the multilayer Gibbs runs.

- **Gaussian mixtures (Tables 4–7, 10).**
  - *Data.* $n/2$ points around each of $\pm\sqrt{\Delta^2/2}\,e_1$, with $N(0, I_p)$ noise, so that $\|\mu_1 - \mu_2\|^2 = 2\Delta^2$.
  - *Model.* Conjugate prior $\mu \sim N(0, \tau^2 I_p)$ with $\tau^2 = 1$, and a DP(1) partition prior.
  - *Samplers.* (S1) is Neal's Algorithm 3 with systematic random-order sweeps. (S2) is Jain & Neal's conjugate split–merge, with 5 restricted-Gibbs launch scans and two proposals per sweep.
  - *Criterion.* "Found" means a second block of size $\geq n/4$, and for the over-clustered starts, two blocks of size ≥ 190 with the largest at least 97% pure. In the other runs, "truth" is inferred from final block sizes of 200 and 200, except in the all-singletons runs with $\Delta^2 \leq 150$, where the composition of both blocks was checked.
  - *Profiles.* These are exact log-posterior differences along a greedy path. At each step the hidden point whose move most increases the posterior is moved. A greedy path is *one* path. Its depth estimates the barrier for building the hidden cluster from its own points, but it does not prove that no lower path exists.
  - *Seeds.* 1–6.
- **Multilayer SBM (Table 8).**
  - *Data.* $n = 400$ and $K_0 = 2$. Each layer is independently assortative or disassortative, with rates $2\rho_n$ and $0.5\rho_n$, where $n\rho_n = 2$.
  - *Model.* Beta(1,1) priors on every block pair and layer, and a DP(1) partition prior.
  - *Computations.* The singleton log-odds are exact, from the one-block state. The greedy path uses exact collapsed log-posteriors.
  - *Sampler.* Single-site collapsed Gibbs with a new-block option, 40 sweeps.
- **Planted-partition SBM (Table 9).**
  - *Data.* $n = 800$, $K_0 \in \{2, 4, 8, 16\}$ equal communities, $p = 0.5$ and $q = 0.05$.
  - *Model.* Beta(1,1) priors on all block pairs and a DP(1) partition prior.
  - *Sampler.* Exact single-site collapsed Gibbs with a new-block option. The incremental bookkeeping was checked against full recomputation of the log-posterior.
  - *Criterion.* "Found" means K_0 blocks of size $\geq n/(2K_0)$. The communities found from one block were counted as distinct majority labels of the large blocks.
  - *Runs.* Seeds 1 and 2, with 30 sweeps from the merged start and 100 from one block.
- **Limitations.** There are few seeds, a single n per model and only the DP prior. In the Gaussian and multilayer examples only $K_0 = 2$ was used. MFM weights change only $O(\log n)$ terms and were not simulated. The point of the simulations is to check the heuristics, not to estimate constants.